The Intramonth Momentum Cycle*

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March 31, 2026

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Abstract

US equity momentum returns accrue almost entirely in just six trading days each month. We show this concentration arises from investors' dash-for-cash: the need to raise cash before month-end payment obligations leads to selling pressure on loser stocks a few days before month-end. A value-weighted WML strategy invested only during days $T-9$ to $T-4$ relative to the last trading day turns \$1 into \$18.8 over 1980–2025, compared to \$2.4 for the rest of the month. The concentration is driven entirely by losers: in stock-level regressions with firm and date fixed effects, bottom-decile losers underperform by an additional 7.2 basis points per day during these six days, while winners show no corresponding pattern. This effect remains stable across subperiods and concentrates among liquid stocks, precisely where institutional liquidity needs would be most visible, suggesting a structural rather than behavioral origin. TAQ data confirm that net selling pressure in losers spikes during the window. We achieve causal identification by exploiting the SEC's May 2024 transition from T+2 to T+1 settlement, which eliminated the mismatch between fund redemption and equity settlement timelines. In a difference-in-differences design, loser underperformance at the former settlement deadline weakens post-reform, shifting toward later days exactly as predicted. All placebo tests return zero. The timing of momentum crash risk further supports this mechanical view: crashes concentrate at month-start, after the monthly payment cycle has cleared. Our evidence points to a simple mechanical driver for the momentum premium: systematic month-end liquidity needs financed by selling loser stocks. These findings, which we replicate across 19 developed markets, suggest that momentum is less about slow information diffusion and more about the fundamental plumbing of the equity market.

JEL Classification: G11, G12, G14, G23

Keywords: Momentum, dash for cash, institutional trading, calendar effects, settlement cycle, market microstructure

*We thank Itamar Drechsler, Amit Goyal, and Dimitri Vayanos for helpful comments, and Filipp Dokienko and Eeli Tuovinen for their excellent research assistance. All errors are our own.

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1 Introduction

Momentum, the tendency of past winners to outperform past losers, is one of the most extensively documented patterns in financial economics. Since Jegadeesh and Titman (1993), hundreds of papers have confirmed momentum profits in equities across dozens of markets and over long historical samples. Two broad classes of explanations have emerged. Behavioral theories attribute momentum to gradual information diffusion or investor underreaction (Hong and Stein, 1999; Daniel et al., 1998). Risk-based theories argue that momentum compensates investors for exposure to crash risk or other priced factors; Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015) document the crash risk and propose strategies to manage it. Despite three decades of research, no consensus has been reached.

We offer a simple "plumbing" explanation for equity momentum in U.S. markets. Rather than asking *why* momentum exists, we ask *when*, and the answer to the second question resolves the first. The bulk of the value-weighted momentum premium in U.S. equities accrues during just six trading days per month. These are roughly the days when, as Etula et al. (2020) document, institutions sell assets to raise cash ahead of month-end settlement. When forced to liquidate, institutions disproportionately sell past losers. This concentrated selling depresses loser prices during a predictable calendar window, mechanically generating the winner-loser spread.

Our analysis begins with daily winner-minus-loser returns constructed using monthly fixed momentum breakpoints from Kenneth French's data library, focusing on 1980–2025. Motivated by the dash-for-cash mechanism documented by Etula et al. (2020), we decompose each month into a six-day pre-month-end window (henceforth "PreTOM") spanning trading days $T-9$ through $T-4$ relative to the last trading day, and the remaining approximately fifteen days. A \$1 investment in 1980 that captures only the window-period WML returns and holds cash otherwise grows to \$18.78 by 2025, compared to \$2.40 for the complementary strategy and \$45.06 for the full WML portfolio (Figure 1).

We establish this through a sequence of evidence: calendar concentration in portfolio

returns, a stock-level loser asymmetry robust to firm and date fixed effects, liquidity concentration among stocks institutions would sell first, TAQ confirmation of selling pressure spikes, a timing shift that follows the 2024 settlement reform one day later, crash risk that concentrates at month-start rather than PreTOM, and replication across 19 developed markets.

The concentration is asymmetric and entirely loser-driven. In portfolio-level returns, losers underperform the market by 9.5 basis points per day during the pre-month-end window ($t = -4.44$), while earning returns indistinguishable from zero on all other trading days. Winners show no differential PreTOM concentration (Figure 2). Momentum profits, in this sense, are not about winners winning. They are about losers losing at predictable times. We confirm the loser-driven window effect in stock-level panel regressions using CRSP data covering over 53 million stock-day observations. We include firm and date fixed effects. Firm fixed effects absorb all time-invariant stock characteristics. Date fixed effects absorb all market-wide variation on any given day, including aggregate risk premia and macroeconomic news. In this more rigorous specification, bottom-decile momentum losers underperform by an additional 7.2 basis points per day (value-weighted) during the six-day window ($t = -3.08$), while winners show no corresponding pattern. The effect is concentrated among liquid losers, exactly those stocks institutions would target when they need to raise cash quickly. To test this liquidity channel, we add a bid-ask spread interaction in equal-weighted regressions, where smaller stocks that drive the BAS variation receive equal weight. The triple interaction Loser $\times$ PreTOM $\times$ BAS is positive and significant (+30.4, $t = 2.57$): the window effect is strongest among stocks with narrow spreads.

The effect is stable across time. Splitting the sample at the midpoint (July 2002), the window effect is significant in both halves: -5.8 bps ($t = -2.31$) in 1980–2002 and -8.5 bps ($t = -2.20$) in 2002–2025. If anything, it has grown stronger as institutional assets have increased. This contrasts sharply with the well-documented attenuation of overall U.S. momentum (Bhattacharya et al., 2017; Jegadeesh and Titman, 2023): the non-window

component has weakened, but the window component has not. The persistence is consistent with a structural friction rather than a behavioral anomaly that arbitrageurs would erode (McLean and Pontiff, 2016).

TAQ data confirm the selling pressure directly. Net selling pressure in losers spikes during the pre-month-end window (Loser × PreTOM = +0.0020, $t = 2.50$). We link this selling pressure PreTOM to investors' liquidity needs. First, using Morningstar data on daily mutual fund flows, we document that during the PreTOM period loser decile stocks experience the largest selling pressure due to mutual fund outflows, while winner portfolio stocks in fact experience on average a small buying pressure instead. Second, we document that loser portfolio stocks have lower dividend yields, on average, compared to other stocks (88% of them in fact pay no dividends) and argue that this leads to higher selling of loser stocks compared to other stocks PreTOM, due to investors' month-end liquidity needs. Evidencing the role of dividends in PreTOM selling, we show that quarterly dividend paying loser stocks do not underperform in the PreTOM period following the payment of the quarterly dividend, but underperform during PreTOM in a statistically significant manner in the month that follows. Behavioral factors are also likely to make investors more prone to sell losers than winners when facing liquidity needs (Akepanidtaworn et al., 2023).[1]

The settlement reform provides a natural experiment of the mechanism. The SEC's May 2024 settlement rule reform brought two important changes relating to end-of-month liquidity-related selling. First, the reform shortened the settlement window from T+2 to T+1. Perhaps more importantly, it eliminated the previously prevailing mismatch between the settlement window that mutual funds offer to their customers (T+1) and the equity settlement (T+2 before the reform). This implies that after the reform, funds anticipating redemptions during PreTOM no longer need to hoard as much liquidity on their balance sheet ahead of time. The net effect is that investors' end-of-month liquidity-related selling

[1] Hartzmark and Solomon (2025) document large price pressures in the stock market from dividend reinvesting. Coval and Stafford (2007) were among the first to document price pressures from mutual fund flows.

shifts closer to month-end post-reform. If the intramonth pattern reflects institutions pre-funding liquidity needs, the timing of selling pressure should shift forward with this change in settlement conventions. This is what we find. In a difference-in-differences analysis comparing $T-4$ returns against $T-3$ returns before and after the reform, the interaction term Post-reform $\times$ $T-4$ is +65.7 basis points ($t = 2.04$), indicating that $T-4$ losers no longer underperform as much under the new settlement regime. The interaction Post-reform $\times$ $T-3$ has the predicted negative sign (-18.2, $t = -0.57$), consistent with selling pressure shifting one day later, though not individually significant. The result strengthens in stock-level panel regressions with firm and date fixed effects: the Loser $\times$ $T-4$ $\times$ Post-T+1 coefficient is +161.4 basis points ($t = 2.45$). Falsification tests using placebo days and dates all return zero.

Besides finding that PreTOM momentum performs well because of the selling pressure on losers, we find that momentum crashes occur mostly in the beginning of the month, when the monthly payment cycle has cleared and when institutions commonly rebalance their portfolios (Harvey et al., 2025). Month-start ($T+1$ to $T+3$) accounts for 20.4% of crash days (WML < -200 basis points), much higher than the 14.3% expected share if crash dates were distributed evenly ($z = 4.87$).[2]

The COVID crash of February–May 2020 is particularly instructive in highlighting the role of investors' liquidity needs, and their desire to sell losers to obtain that liquidity, in driving momentum returns. During the COVID period, PreTOM selling pressure on losers was an order of magnitude larger than normal. Over the four-month episode, the pre-month-end window WML earned positive cumulative returns of +1,424 bps. This partly reversed during month-start, with WML recording a $-9{,}515$ basis point cumulative loss (the momentum crash). This result contributes to the momentum crash literature that argues crashes occur predominantly after long bear markets (Daniel and Moskowitz, 2016), which was not the case for COVID.

[2]The result is robust across thresholds. At -100 bps: 31.8% vs. 33.3% ($z = -1.26$, $p = 0.21$). At -300 bps: 29.9% vs. 33.3% ($z = -1.24$, $p = 0.22$). PreTOM crash frequency is proportional at every threshold.

The calendar concentration of momentum profits poses distinct challenges to the two dominant classes of momentum explanations. Theories of momentum based on gradual information diffusion, underreaction, or overconfidence (Hong and Stein, 1999; Daniel et al., 1998; Andrei and Cujean, 2017) predict that momentum returns should accrue continuously as information is slowly incorporated into prices or as uncertainty slowly dissipates. Information does not arrive in line with the settlement schedule. The fact that the bulk of the premium concentrates in six specific days tied to institutional and retail investors' cash management, and is absent on the remaining fifteen, is also difficult to reconcile with models in which momentum reflects the speed of belief updating. Risk-based theories face a different problem. If momentum compensates investors for bearing higher risk in momentum winners as in Cochrane et al. (2008), or exposure to a priced factor, the returns should accrue evenly and not during the PreTOM window. The fact that crash risk concentrates in a different calendar window ($T+1$ to $T+3$) than profits ($T-9$ to $T-4$) is also difficult to explain in this framework.

Several additional findings complete the picture. The window effect is present every month, not just at quarter-ends, ruling out mutual fund window dressing (Brown, 2017) as the primary explanation. In December, the PreTOM effect weakens, because tax-loss selling depletes the pool of losers already by mid-month, before the pre-month-end window selling begins. The effect is also robust to restricting the sample to S&P 500 constituents. Finally, the PreTOM loser effect extends beyond U.S. equities: across 19 developed markets, momentum losers underperform the local market by an additional 4.2 basis points per day during the PreTOM window ($t = -4.60$), while winners show no corresponding pattern, consistent with the purely loser-driven signature we document in the U.S. The effect is absent in Japan, where also momentum returns are approximately zero (Fama and French, 2012; Asness et al., 2013).

Our results are most compatible with flow-based theories of momentum, as depicted in Lou (2012) and Vayanos and Woolley (2013). They emphasize the role of performance-

driven flows to delegated asset managers: past returns generating investor inflows that induce managers to trade in the direction of prior performance, producing momentum. Our flow-based evidence is consistent with this, but the selling of losers that we document can also be rationalized by investors' liquidity needs (as losers pay lower dividends) or behavioral trading patterns that lead to selling of losers (Akepanidtaworn et al., 2023).[3]

We also contribute to the debate over whether momentum has weakened (Jegadeesh and Titman, 2023; Bhattacharya et al., 2017; Chordia et al., 2014). Our decomposition shows that the non-window component of momentum has indeed attenuated, while the window component persists. This is consistent with a structural institutional-mechanics origin rather than a behavioral anomaly that arbitrage would erode.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes our data and empirical design. Section 4 presents the main results. Section 5 provides direct evidence on the mechanism, including TAQ selling pressure, mutual fund flows, the channels through which losers bear disproportionate selling, the T+1 settlement experiment, and crash risk separation. Section 6 concludes.

2 Related Literature

The momentum premium was first documented by Jegadeesh and Titman (1993) and it has been confirmed across international markets (Rouwenhorst, 1998; Griffin et al., 2003; Asness et al., 2013) and in long historical samples (Geczy and Samonov, 2016). It has resisted a unified explanation. Behavioral theories predict gradual accrual as information slowly diffuses (Hong and Stein, 1999; Barberis et al., 1998; Daniel et al., 1998; Andrei and Cujean, 2017); and risk-based theories predict permanent compensation for crash exposure (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015). Neither predicts that the premium

[3] Vayanos and Woolley (2013) present a theory that can rationalize why predictable flow-driven price movements are not fully arbitraged away ahead of time. This is in line with our results as well as the evidence in Etula et al. (2020) and Hartzmark and Solomon (2025) who show that predictable flows lead to price effects also at the moment when the flows take place.

should concentrate in six settlement-linked days. Recent research has argued that momentum premium has attenuated (Bhattacharya et al., 2017; Chordia et al., 2014; Jegadeesh and Titman, 2023). Our decomposition shows that the non-window component has attenuated, while the $T-9$ to $T-4$ window component, driven by institutional settlement frictions, has not.

Flow-based momentum. Several papers link institutional trading to momentum. Lou (2012) documents at quarterly frequency flow-driven buying and selling by mutual funds and links this to momentum. Vayanos and Woolley (2013) in turn provide a theoretical model that rationalizes flow-driven momentum: performance-driven flows generate positive-feedback trading that produces momentum in equilibrium.[4]

Turn-of-month dash for cash. We also show that momentum is linked to institutional and potentially also retail trading, but point out the relevance of investors' liquidity needs in the genesis of that trading behavior. Etula et al. (2020) document that institutional investors sell stocks before month-end (during our PreTOM period) to meet their turn-of-the-month liquidity needs, generating a predictable dip in aggregate returns followed by a turn-of-month reversal. Our research extends the dash-for-cash framework from the market level to the cross-section: month-end selling is not uniform across stocks but concentrates in past losers, thus causing selling pressure on loser stocks. This selling pressure appears to be a large driving force behind momentum as most momentum profits accrue during the six-day PreTOM period. The T+1 natural experiment provides causal identification.

Why do investors sell specifically losers and withdraw money from mutual funds holding loser stocks to obtain their end-of-month liquidity? This can be related to their updating of information—as argued by Vayanos and Woolley (2013)—but there is also a simpler explanation consistent with the timing of the sales: owners of loser stocks sell before month-

[4] Grinblatt et al. (1995) document that mutual funds are momentum traders. Sias (2004) shows that institutional herding contributes to momentum. Puckett and Yan (2011) confirm momentum-chasing behavior in daily institutional trading data.

end as they too need end-of-month liquidity to cover payments due at month-end (according to Etula et al. 2020, the payment cycle clears largely at month-end). The finding that the selling concentrates on loser stocks can be linked to lower dividends of the loser stocks (88% of them do not pay dividends at all). We show that dividends matter using data on the subset of loser stocks that pay quarterly dividends: such stocks in the loser portfolio are less likely to be sold in the PreTOM period that follows quarterly dividend payments. Behavioral reasons are also likely to matter: Akepanidtaworn et al. (2023) show that institutional investors sell in a heuristic, non-strategic manner, reaching for the most salient positions under time pressure. Finally, taxes are likely to matter: past losers, which have already declined and carry unrealized losses that can offset gains (Constantinides, 1984; Grinblatt and Moskowitz, 2004), are the path of least resistance. In addition, mutual funds' window dressing may play a role in deciding which stocks to sell before month-end (Brown, 2017). Potentially all of these reasons are important in explaining why there is abnormally high selling of loser stocks during the PreTOM period.

Market plumbing. Our paper contributes to a growing literature showing that institutional frictions generate asset pricing phenomena previously attributed to behavioral or risk-based stories. Duffie (2010) argues that slow-moving capital explains many apparent anomalies. He and Krishnamurthy (2013) show that intermediary balance sheet constraints affect asset prices through funding channels. Du et al. (2018) demonstrate that CIP deviations arise from settlement and funding frictions. Kutai et al. (2025) provide direct evidence that settlement mechanics shape market outcomes during stress. U.S. equity settlement has shortened progressively: from T+3 (1993) to T+2 (September 2017) to T+1 (May 2024).

Our paper is closely related also to Hartzmark and Solomon (2025), who show that predictable dividend reinvestment flows generate aggregate market price pressure across 58 international markets; our paper extends this logic to the cross-section, showing that month-end liquidity flows generate disproportionate selling pressure on momentum losers.

We exploit the T+2→T+1 transition as a natural experiment; to our knowledge, this is the first paper to use settlement-cycle changes for asset pricing identification.

3 Data and Methodology

3.1 Data Sources

We use three primary data sources. First, for both portfolio-level and stock-level analysis, we construct daily momentum decile portfolios from CRSP using fixed monthly sorts: all NYSE, AMEX, and NASDAQ stocks are ranked into deciles based on cumulative returns over months -12 through -1, skipping the most recent month, using NYSE breakpoints. Decile assignments are held constant within each calendar month. This follows the standard construction of Fama and French (1996) rather than the original Jegadeesh and Titman (1993) implementation, which uses all-stock breakpoints and equal-weighting; the Fama-French convention has been adopted by the vast majority of the post-JT empirical literature. From Kenneth French's data library we obtain daily Fama-French factors, including the market excess return ($r_t^m - r_t^f$) and the risk-free rate (r_t^f, the one-month Treasury bill rate).[5] Our portfolio-level sample spans 1980–2025.

Second, for stock-level analysis, we construct a panel of daily stock returns from CRSP merged with momentum decile assignments and bid-ask quotes. The stock-level panel contains approximately 53.3 million stock-day observations spanning 1980–2025, with momentum deciles assigned using fixed monthly sorts (held constant within each calendar month).

Third, for direct evidence on institutional trading, we obtain daily measures of selling pressure from the WRDS Intraday Indicators dataset, derived from NYSE TAQ data and covering 2003–2022.[6] These include net selling pressure and the sell share of volume. We merge these measures with the CRSP panel at the stock-day level.

[5] Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.
[6] See https://wrds-www.wharton.upenn.edu/pages/grid-items/wrds-intraday-indicators/.

Fourth, for evidence on mutual fund flow pressure, we use daily fund flows from the Morningstar database matched with quarterly fund holdings from the CRSP Mutual Fund Holdings database. We calculate stock-specific flow pressure by allocating each fund's daily flow to individual stocks in proportion to the fund's prior-quarter holdings, normalized by the stock's previous-day market capitalization. The matched sample covers approximately 30% of all mutual fund total net assets in CRSP and Morningstar, spanning 2010–2023.

For international evidence (Section 5.5), we construct analogous daily momentum decile portfolios for 19 developed markets using Compustat Global. We apply size-conditional breakpoints within each country, analogous to NYSE breakpoints in the U.S. analysis, and use fixed monthly sorts; this differs from the Kenneth French international momentum portfolios, which use tercile sorts with daily rebalancing and are therefore less comparable to our U.S. baseline.

3.2 Key Variables

Trading day index. For each month, we index trading days backward from the last trading day. The last trading day is $T = 0$, the penultimate is $T = -1$, and so on. Our pre-month-end window ("PreTOM") consists of days $T = -9$ through $T = -4$, corresponding to the second-to-last week of the month. This window aligns with the period of elevated institutional selling documented by Etula et al. (2020).

Momentum portfolios. The winner portfolio consists of stocks in the top momentum decile; the loser portfolio consists of stocks in the bottom decile. The WML (winner-minus-loser) return is the daily difference. We examine both value-weighted (VW) and equal-weighted (EW) portfolio returns.

Loser indicator. In stock-level regressions, $\text{Loser}_{i,d}$ is an indicator equal to one if stock i is in the bottom momentum decile on date d. The variable PreTOM_d indicates whether date d falls in the $[T-9, T-4]$ window, where t denotes the within-month event-time index defined

above. The interaction $\text{Loser}_{i,d} \times \text{PreTOM}_d$ is our key variable, capturing the additional return earned by losers during the window.

Bid-ask spread. From CRSP daily bid and ask quotes, we compute the percentage bid-ask spread as $\text{BAS}_{i,d} = (A_{i,d} - B_{i,d})/M_{i,d}$, where A is the ask price, B is the bid price, and M is the midpoint.

Selling pressure. From TAQ, we construct: (i) net selling pressure, defined as the share of total dollar volume initiated by sellers minus the share initiated by buyers; and (ii) sell share of volume, the fraction of volume from seller-initiated trades.

Settlement transitions. If precautionary selling is driven by the settlement mismatch between fund redemptions (T+1) and equity settlement, exogenous changes to the settlement cycle should shift the selling window. U.S. equity settlement shortened from T+3 to T+2 (September 5, 2017) and from T+2 to T+1 (May 28, 2024).[7] The May 2024 reform eliminated the mismatch entirely by aligning equity settlement with fund redemption settlement at T+1 (see Section 5.3). We exploit this transition as a natural experiment using a unified DiD that compares three adjacent trading days, $T-4$, $T-3$, and $T-2$, before and after the reform, with $T-2$ as the control. We define PostT1_d as an indicator for dates on or after May 28, 2024. The earlier T+3→T+2 transition, which did not eliminate the mismatch, serves as a falsification test.

3.3 Empirical Design

Our stock panel regressions take the form:

$$r_{i,d} - r_d^f = \alpha_i + \delta_d + \beta_1 \,\text{Loser}_{i,d} + \beta_2 \,\text{Loser}_{i,d} \times \text{PreTOM}_d + \varepsilon_{i,d} \tag{1}$$

[7] The T+3 to T+2 transition was mandated by SEC Rule 15c6-1(a) (Securities and Exchange Commission, 2017). The T+2 to T+1 transition was mandated by Securities and Exchange Commission (2023).

where d indexes trading dates, α_i and δ_d are firm and date fixed effects, and standard errors are double-clustered by firm and date. We estimate both equal-weighted (EW) and value-weighted (VW) specifications. In the VW specification, observations are weighted by lagged market capitalization within each momentum decile-date cell, so that the regression coefficients reflect the returns of portfolios that overweight larger, more liquid stocks.

4 Main Results

4.1 The Window Effect in Portfolio Returns

We begin with the portfolio-level evidence. Figure 2 plots average daily market-adjusted returns for momentum winners, losers, and the WML spread during the pre-month-end window ($T-9$ to $T-4$) versus the rest of the month. The WML spread is large and positive during the pre-month-end window and close to zero on all other trading days, with the difference driven almost entirely by losers rather than winners.

To illustrate the economic magnitude of this concentration, Figure 1 decomposes the momentum premium into window and non-window components. Each day's value-weighted WML return (top decile minus bottom decile, using fixed monthly decile assignments from our CRSP panel) is assigned to one of two mutually exclusive buckets based on the trading-day index t: the "window" return on days $T-9$ through $T-4$, and the "rest" return on all other days. Compounding each component separately from \$1 on January 2, 1980 reveals a stark divergence. By 2025, the window-only strategy has grown to \$18.8 while the rest-only strategy reaches just \$2.4.[8] The two strategies track each other through the early 2000s before diverging, consistent with the non-window component of momentum weakening over time while the window component persists.

The asymmetry is entirely loser-driven. Losers underperform the market by 9.5 basis

[8] Using French's daily-rebalanced momentum portfolios yields \$15.9 and \$2.2 respectively. Daily rebalancing removes recovering stocks from the loser decile each day, which slightly reduces the window return and attenuates the rest-of-month return. The qualitative pattern is identical.

points per day during the pre-month-end window ($t = -4.44$) but earn returns indistinguishable from zero on all other trading days (-2.2 bps/day, $t = -1.45$). Figure 2 confirms this visually: the loser bar is large and negative during PreTOM, while winners show no special pre-month-end pattern. Momentum, at the portfolio level, is about losers losing during six predictable days each month. The pattern is robust across subperiods: Appendix Figures A2 and A3 show that the loser PreTOM underperformance and the WML window concentration are present in both the 1980–2002 and 2002–2025 halves of the sample. Appendix Figure A1 provides the full daily decomposition, plotting mean excess returns for each trading day T separately for losers and winners.

4.2 Stock-Level Evidence

The portfolio results establish that the concentration is driven by losers, not winners. To quantify the effect at the stock level and rule out a wide range of confounds, we estimate panel regressions with firm and date fixed effects. The firm fixed effects absorb any time-invariant stock characteristics, such as size, illiquidity premia, or persistent return differences across stocks. The date fixed effects absorb all market-wide variation on a given day, including aggregate risk premia, macroeconomic news, and any calendar anomaly affecting all stocks equally.

Table 1 presents estimates of equation (1). In the value-weighted specification, the coefficient on Loser $\times$ PreTOM is -7.2 basis points per day ($t = -3.08$). Over six trading days, this implies losers underperform by approximately 43 basis points per month solely during the window, a magnitude sufficient to account for the entire monthly momentum premium. The equal-weighted estimate is smaller (-2.6 bps, $t = -1.85$) because equal-weighting dilutes the effect across illiquid micro-cap stocks that do not participate in institutional selling, a point we clarify below with bid-ask spread interactions.

4.2.1 Liquidity

If the window effect is driven by institutional selling, it should concentrate where institutions trade: among liquid stocks.

Table 2 adds the contemporaneous bid-ask spread and its interactions. In the equal-weighted specification, the triple interaction Loser × PreTOM × BAS is large and positive (30.4, $t = 2.57$), indicating that the window effect concentrates among liquid losers: stocks with narrow spreads show large pre-month-end declines, while illiquid stocks exhibit little selling pressure. This is precisely what the dash-for-cash hypothesis predicts: when institutions need to raise cash quickly, they sell stocks they *can* sell, not stocks that happen to be cheap. The value-weighted interaction is also positive and significant (+58.8, $t = 2.34$), confirming the liquidity channel holds regardless of weighting scheme.

Since value-weighting naturally captures the institutional-trading channel, we adopt it as our primary specification for all remaining analyses.

4.2.2 Subperiod Stability

A natural concern is that the window effect reflects an artifact of earlier, less liquid markets, such as thin trading, wider spreads, and less institutional participation, that has since disappeared as markets modernized. Many well-documented anomalies do attenuate over time as arbitrage capital grows and markets become more efficient (McLean and Pontiff, 2016; Greenwood and Sammon, 2025). If so, the window effect should be stronger in the first half of the sample and attenuate or vanish in the second. The opposite is true.

Table 3 splits the sample at the midpoint (July 2002) into two equal halves. The window effect is present in both: −5.8 bps ($t = -2.31$) in the first half and −8.5 bps ($t = -2.20$) in the second. The increase in magnitude is consistent with growing institutional assets amplifying month-end selling pressure over time. An illiquidity explanation predicts the opposite pattern: if thin, less-efficient pre-2002 markets drove the effect, it should attenuate as spreads tightened and markets modernized (McLean and Pontiff, 2016). The strengthening

rules this out. The Loser main effect, which captures average daily underperformance outside the window, roughly doubles from the first half to the second, yet the window component is significant throughout. The well-documented attenuation of U.S. momentum (Bhattacharya et al., 2017; Jegadeesh and Titman, 2023) appears to reflect erosion of its non-window component, while the institutional-mechanics component persists. Unlike anomalies driven by mispricing, the window effect reflects a structural "plumbing" friction that is not easily arbitraged away: institutions face recurring, non-discretionary cash needs at month-end regardless of whether arbitrageurs are aware of the pattern or positioned to trade against it. Internet Appendix Table IA.1 shows that loser PreTOM underperformance extends back to 1927, confirming the pattern is not an artifact of the post-1980 institutional era.

A related concern is that the persistence of PreTOM profits reflects compensation for crash risk. If momentum crashes disproportionately fell in the pre-month-end window, the strategy's apparent profitability would simply be a tail-risk premium. We address this directly in Section 5.4: crash days fall in the PreTOM window at roughly the rate expected under a uniform distribution, neither more nor less than in other non-month-start windows. What is structurally unusual is month-start, where crash days are overrepresented by 43%.

The PreTOM window $[T-9, T-4]$ is the unique trough in loser returns across the second half of the trading month.[9]

5 Mechanism

5.1 TAQ Selling Pressure and Mutual Fund Flows

If the window effect reflects institutional selling of losers, we should observe elevated selling pressure in loser stocks during the PreTOM $[T-9, T-4]$ window. We test this using intraday data from the WRDS TAQ Intraday Indicators dataset, which provides daily measures of

[9] A sliding six-day window analysis confirms that mean market-adjusted loser returns decline monotonically from near zero at mid-month to a trough at PreTOM (-7.7 bps/day, $t = -3.59$, Newey-West), before partially recovering at month-end (Online Appendix).

buy- and sell-initiated volume classified by the Lee-Ready algorithm. The sample covers 2003–2022, yielding approximately 17.9 million stock-day observations.

Table 6 regresses selling pressure on Loser, PreTOM, and their interaction, with firm and date fixed effects. The Loser × PreTOM coefficient on net selling pressure is 0.0020 ($t = 2.50$), and on the sell share of volume 0.0010 ($t = 2.50$). When a stock enters the bottom momentum decile, it experiences more selling during the pre-month-end window than the same stock does outside the window or when it is not classified as a loser.[10]

Our next piece of evidence on the forces behind PreTOM price pressures comes from mutual fund flows. We calculate stock-specific flow pressures from mutual funds' daily flows assuming all funds invest their flow into stocks (or sell stocks in case of outflows) in proportion to their holdings at the end of the previous quarter. The flow pressure measure is the sum of such expected flow-related buying and selling normalized by the stocks' market capitalizations prevailing one day before the reported flows. Figure 6 documents the average mutual fund flow pressures on stocks in each momentum decile during the PreTOM window. The pattern is monotonic: losers (decile 1) face the strongest outflow pressure (−9.9), while winners (decile 10) experience slight inflow pressure (+0.5).

It seems investors also withdraw funds during PreTOM disproportionately from mutual funds that hold loser stocks. We next examine possible reasons for this.

5.2 Why Losers Face Selling Pressure During PreTOM

The TAQ evidence confirms that losers face elevated selling pressure during PreTOM. This selling of losers is linked to withdrawal of funds from mutual funds holding loser stocks. But why do investors disproportionately sell momentum loser stocks or withdraw funds from mutual funds holding loser stocks when they need cash? We offer three complementary

[10] The 0.0020 coefficient represents an additional 0.2 percentage points of net selling pressure on a $[-1, 1]$ scale, or 16% of the Loser main effect (0.0122). This is a lower bound. Lee-Ready classification is approximately 85% accurate, and misclassification biases order imbalance measures toward zero (Odders-White, 2000). Moreover, TAQ captures only on-exchange trades; approximately 40–45% of U.S. equity volume now executes off-exchange, where institutional block orders preferentially route.

explanations: lower dividend yield among loser stocks, tax-related selling of losers, and behavioral selling patterns that make institutions sell losers during the PreTOM window.

5.2.1 Dividend Income

A first reason why losers bear disproportionate PreTOM selling is that they generate less dividend income. Panel A of Table 8 shows the composition difference: 88.4% of D1 loser stock-months are non-payers, compared to 72.0% for D10 winners. Only 8.7% of losers pay quarterly dividends, and thus offer no income during eight months of the year. Investors holding losers therefore receive less cash from dividends and are more likely to need to sell shares to meet month-end liquidity needs.

If dividend income provides a cash buffer that reduces the need to sell, the PreTOM effect among quarterly dividend payers should vary with how recently the dividend was paid. We test this by splitting months within the quarterly payment cycle: Month 1 is the month in which the dividend was paid (or is paid before month-end), Month 2 is one month after payment, and Month 3 is the month before the next payment. In Month 1, the investor has cash in hand from the recent dividend (or expects to receive it by month-end); by Month 2, that cash has been spent or allocated. If investors sell losers to obtain month-end liquidity, then stocks paying quarterly dividends should not underperform during PreTOM in Month 1.

Panel B of Table 8 reports value-weighted stock-level regressions restricted to quarterly payers in D1 (losers) and D10 (winners). The PreTOM effect is weakest in Month 1, when dividend cash is available (-6.3 bps, $t = -1.32$, insignificant), and strongest in Month 2, when the cash buffer has been depleted (-10.6 bps, $t = -1.98$). Panel C confirms the pattern in TAQ data: relative to Month 1, net selling pressure during PreTOM is significantly higher in Month 2 ($+0.0038$, $t = 4.99$) and Month 3 ($+0.0018$, $t = 2.34$). The pattern traces the depletion of dividend income within the quarter: as the cash buffer drains, forced selling during PreTOM intensifies.

5.2.2 Tax-Related Selling

A natural alternative is that the pre-month-end selling of losers reflects tax-motivated trading rather than institutional liquidity needs. Grinblatt and Moskowitz (2004) show that the capital gains "overhang" is a key predictor of momentum returns, with the loser momentum effect roughly 2.5 times larger in December than in other months. Constantinides (1984) makes clear that optimal tax-loss harvesting should take place each month during the year. Hence generating tax losses can be one reason why investors prioritize selling loser stocks during PreTOM.

We have mixed evidence related to tax-loss harvesting. First, tax-loss harvesting should amplify the PreTOM effect in December by intensifying selling of losers before year-end (the last opportunity to generate losses during that calendar year). This we do not observe: PreTOM returns are higher in December compared to other months. Loser PreTOM returns in December average +0.18% per month ($t = 0.28$), indistinguishable from zero, compared to −0.67% across all months ($t = -3.25$). By comparison, September, where tax-loss incentives are weaker, produces the deepest underperformance in the sample (−2.37%, $t = -3.36$).

There is, however, a tax-related explanation for this pattern which shows that tax considerations also matter for loser selling: mutual fund capital gains distribution record dates cluster in mid-December.[11] Investors who learn estimated distribution sizes in November are likely to harvest offsetting tax losses in the first two weeks of December, well before our PreTOM window opens around December 17–26. By the time PreTOM begins, much of the selling pressure on losers has already occurred. This mechanism predicts attenuation of the PreTOM effect in December, and that is exactly what we find.

Our finding complements Grinblatt and Moskowitz (2004). Tax-loss selling amplifies December's monthly loser underperformance, but this selling occurs earlier in the month, front-running the institutional selling that would normally concentrate in PreTOM (Ap-

[11] For example, 65% of Vanguard's 2023 year-end distribution record dates fell between December 14 and 19 (Vanguard, *Final Estimated Year-End Capital Gains Distributions*, December 2023, available at https://investor.vanguard.com).

pendix Figure A4). December illustrates this displacement clearly. Despite being among the stronger months for momentum at the monthly frequency (+137 bps, ranking fifth in our sample), December produces near-zero PreTOM underperformance. September presents the mirror image: the deepest PreTOM underperformance in the sample coincides with strong total momentum, consistent with the liquidity-related mechanism operating without seasonal disruption.

This is not a contradiction: December demonstrates that loser selling can generate a strong monthly premium even when displaced from the PreTOM window. The settlement cycle determines the timing in normal months but tax deadlines override that timing in December. Both channels route through the same friction, selective institutional liquidation of past losers, and both generate momentum. The mechanism is the selling itself, not the specific days.

5.2.3 Behavioral Selling

When fund managers need to raise cash under time pressure, they may reach first for the most salient positions: stocks that are actively falling *now*, rather than those that fell months ago and have since stabilized. This "fresh pain" heuristic is consistent with the behavioral selling patterns documented by Akepanidtaworn et al. (2023), who show that institutional investors make faster and less deliberate decisions when selling than when buying.

We test this within the bottom momentum decile by measuring "freshness": the cumulative return over the most recent three months ($\text{ret}_{3,1}$), standardized cross-sectionally within D1 each month and sign-flipped so that higher values indicate worse recent performance. Crucially, all stocks in D1 have similar 12-month losses (the sorting variable), so freshness captures the *timing* of the loss, not its magnitude. Fresh losers (bottom tercile of $\text{ret}_{3,1}$, mean $= -33.8\%$) and stale losers (top tercile, mean $= +14.9\%$) have nearly identical 12-month returns (-56.3% vs. -45.2%).

Table 9 reports the results. Panel A splits D1 into terciles. With firm and date fixed

effects, bid-ask spread controls, and double-clustered standard errors, fresh losers underperform by -3.3 basis points per day during PreTOM ($t = -2.17$), while stale losers show no window effect ($+1.2$ bps, $t = 1.02$). The difference is significant ($F = 7.15$, $p = 0.008$): fresh losers underperform stale losers by 4.5 bps per day during the six-day window. Panel B uses a continuous freshness measure. The PreTOM $\times$ freshness effect is -1.8 bps per standard deviation ($t = -2.38$), strengthening after controlling for bid-ask spreads (from -1.5 without controls). The effect is not driven by liquidity.

The freshness effect vanishes in December: the December $\times$ PreTOM $\times$ freshness interaction is $+3.8$ ($t = 1.61$), leaving the total December effect at $+1.7$ ($t = 0.74$, insignificant). Tax-loss selling depletes fresh losers before PreTOM begins, eliminating the effect. Outside December, the effect strengthens to -2.1 ($t = -2.72$).

5.2.4 Window Dressing

Finally, we consider window dressing as a reason for poor PreTOM returns for losers. Brown (2017) documents that momentum returns concentrate in the third month of calendar quarters, attributing the pattern to tax-loss selling and window dressing by delegated managers.

If window dressing is the primary driver of the pre-month-end loser effect, the pattern should amplify in December and at quarter-ends. Our evidence points in the opposite direction. The PreTOM effect is present every month, with no quarter-end amplification, and is sharply attenuated in December (Appendix Figure A4).

Two tests address the window dressing alternative directly. First, restricting the sample to non-quarter-end months, where window dressing is least relevant, the Loser $\times$ PreTOM coefficient is -7.5 bps ($t = -2.66$), virtually identical to the full-sample estimate (Table 5). Second, interacting Loser $\times$ PreTOM with a quarter-end indicator produces a small, insignificant triple interaction (Table A1): there is no systematic quarter-end amplification, consistent with a monthly liquidity cycle rather than quarterly reporting incentives.

5.3 T+1 Settlement Natural Experiment

The preceding evidence establishes a strong association between the institutional settlement calendar and the timing of momentum returns, but it is ultimately correlational. An alternative explanation could attribute the calendar concentration to any monthly cycle, whether behavioral, informational, or macroeconomic, that happens to coincide with the pre-month-end window. To move toward a causal interpretation, we exploit a regulatory change that shifted the precautionary selling window by exactly one day: the SEC's May 28, 2024 transition from T+2 to T+1 equity settlement.

Prior to the 2024 reform, open-end mutual funds faced a settlement mismatch: fund redemptions required cash payment on a T+1 basis while sales of portfolio securities settled on T+2 (Securities and Exchange Commission, 2023). To meet redemption payments on time, funds were compelled to sell portfolio securities one day earlier than mutual fund settlement requires. This mismatch forced mutual funds to make precautionary asset sales in anticipation of month-end withdrawals from investors who needed cash at $T-1$ to make payments at T. As a result, under the T+3 settlement regime that prevailed until September 2017, mutual funds were selling stocks in a precautionary manner to obtain end-of-month liquidity during the PreTOM period, with selling being largest from $T-6$ to $T-3$ morning (Etula et al. 2020, whose sample predates the 2017 reform), and by $T-3$ afternoon the selling pressure was already over. The T+2 regime (2017–2024) shortened settlement by one day but preserved the mismatch, as fund redemptions continued to settle on T+1 while equities settled on T+2.

In September 2017 the settlement period was shortened to T+2 and in May 2024 to T+1. In addition, and potentially most importantly, in May 2024 the one-day settlement finally matched the settlement window that mutual funds offer to their customers, eliminating the need for mutual funds' precautionary selling to meet their end-of-month liquidity demand. Both the SEC and DTCC characterize the transition as a major change in U.S. market infrastructure, requiring coordinated adjustments across trading, clearing, and funding sys-

tems (Securities and Exchange Commission, 2023; Depository Trust & Clearing Corporation, 2024). With no need to pre-sell, and with a one-day settlement window both for mutual funds and the stock market post 2024, the key prediction concerns day $T-4$, which we empirically identify as the last day of elevated precautionary selling in our full sample (Appendix Figure A1). Once the mismatch is eliminated, funds no longer need to complete their selling by $T-4$. That day should revert to normal, with any remaining selling pressure shifting to $T-3$ or later. We find strong support for this.

In fact, the pattern of poor PreTOM returns in the window $T-9$ to $T-4$ that held for 45 years appears to vanish after May 2024 when mutual funds' need for precautionary selling disappears. In the 533 months before the transition, compounded loser returns during the PreTOM window ($T-9$ to $T-4$) averaged -46.5 basis points per month ($t = -3.29$). In the 19 months after, the same window averages $+12.5$ basis points ($t = 0.36$, insignificant). However, with only 19 post-event months, this could reflect noise rather than a structural break.

The day-by-day return profiles paint a clearer picture. Figure 3 plots the difference in mean loser market-adjusted returns (Post-T+1 minus Pre-T+1) at each trading day T. Following the new settlement rule in May 2024, $T-3$ absorbs -27 basis points of additional selling pressure, exactly as predicted. Day $T-4$ shifts up by $+43$ basis points, suggesting institutions abandoned the old precautionary selling practices rapidly after the reform. Day $T-2$ also turns slightly negative, consistent with some selling spilling beyond $T-3$. All other days show differences indistinguishable from zero. The PreTOM pattern did not disappear. It shifted.

The day-level evidence is visually compelling but faces an obvious confounder: any change in macroeconomic conditions or momentum portfolio behavior after May 2024 could drive the result. We design a DiD that isolates a precise prediction of the settlement mechanism. The test compares $T-4$ and $T-3$, the two days at the boundary of the old and new selling windows, against early-month trading days ($T+5$ through $T+8$) that are unrelated to settle-

ment mechanics. Days closer to month-end, such as $T-2$ or $T-1$, could be contaminated by the reversal from precautionary selling and are therefore less suitable as controls.[12] Under T+1, day $T-4$ should lose its special status as precautionary selling is no longer necessary that early, while $T-3$ should absorb whatever selling pressure remains. We estimate:

$$r_d^{\text{loser}} - r_d^{\text{mkt}} = \alpha + \beta_1 \, \mathbf{1}[T{=}{-4}]_d + \beta_2 \, \mathbf{1}[T{=}{-3}]_d + \beta_3 \, \text{PostT1}_d + \beta_4 \, \mathbf{1}[T{=}{-4}]_d \times \text{PostT1}_d + \beta_5 \, \mathbf{1}[T{=}{-3}]_d \times \text{PostT1}_d + \varepsilon_d \quad (2)$$

where $\mathbf{1}[T{=}{-4}]$ indicates the day that was the last precautionary selling day in our pre-reform sample, $\mathbf{1}[T{=}{-3}]$ indicates the day predicted to absorb selling pressure under T+1, PostT1 indicates dates after May 28, 2024, and early-month days ($T+5$ through $T+8$) serve as the omitted baseline. This specification delivers three testable predictions in a single regression: $\beta_4 > 0$ (the old selling day is relieved), $\beta_5 < 0$ (the new selling day absorbs pressure), and $\beta_4 - \beta_5 > 0$ (the full shift from old to new).

Table 7 reports the results. At the portfolio level with the pre-period restricted to the T+2 era (September 2017–May 2024), $\beta_4 = +65.7$ basis points ($t = 2.04$): day $T-4$ improves relative to early-month days after the reform. The coefficient $\beta_5 = -18.2$ ($t = -0.57$) has the predicted negative sign, though it is not individually significant. The full shift $\beta_4 - \beta_5 = +83.9$ ($t = 2.22$, $p = 0.027$) is significant: the divergence between the old and new selling days is large and precisely estimated.

The portfolio-level DiD exploits time-series variation only: three days per month, 19 post-reform months. We gain substantial power by moving to the stock-level panel (Panel C of Table 7). Using the same three-day specification in the CRSP panel (firm and date fixed effects, value-weighted, standard errors clustered by date), $\beta_4 = +161.4$ basis points ($t = 2.45$), while $\beta_5 = -43.5$ ($t = -0.76$), the right sign though not individually significant. The full shift $\beta_4 - \beta_5 = +205.0$ ($t = 2.54$, $p = 0.011$). On the same day, losers at the

[12] In untabulated results, using $T-2$ as the control instead of early-month days produces quantitatively similar estimates.

settlement boundary improve relative to winners by over 200 basis points after the reform. Three falsification tests confirm the result is specific to the settlement reform: placebo days ($T-6$ vs. $T-7$, $\beta_4 - \beta_5 = -2.3$, $t = -0.08$) and two placebo dates (May 2020: +20.8, $t = 0.79$; May 2018: +11.9, $t = 0.54$) all return zero (Online Appendix).

The earlier T+3→T+2 transition (September 2017) produces no detectable shift in the same unified specification ($\beta_4 - \beta_5 = +0.9$, $t = 0.04$; Online Appendix), using $T-4$ and $T-3$ against the same early-month control and excluding the post-T+1 period. This is exactly what one would expect. The 2017 reform shortened settlement by one day but did not resolve the fundamental problem: funds continued to settle redemptions on T+1 while portfolio securities settled on T+2. The SEC's own subsequent decision to reform settlement again in 2024, explicitly citing the persistent mismatch between fund redemption settlement and equity settlement as the primary motivation (Securities and Exchange Commission, 2023), is direct evidence that the 2017 reform left the friction intact. It took a second reform to close that gap.

5.4 Momentum Crashes

A natural concern is that PreTOM's strong performance reflects luck: momentum crashes happened to fall outside the window by coincidence. We classify a trading day as a "crash day" if VW WML falls below −200 basis points. To avoid confounding PreTOM and month-start, we compare each window against remaining trading days only: PreTOM is compared against non-month-start days, and month-start against non-PreTOM days.[13] PreTOM accounts for 30.9% of crash days versus its 33.3% calendar share ($z = -1.27$, $p = 0.20$): crashes arrive in PreTOM at the expected rate. What *is* unusual is month-start, which accounts for a disproportionate share of crash days at every threshold from −100 to −300 bps (Figure 4).

[13] Calendar share is the fraction of trading days that fall in a given window within the relevant comparison universe. For PreTOM, we exclude month-start days from both numerator and denominator; for month-start, we exclude PreTOM days. If crashes were spread evenly across the month, each window's share of crash days would equal its calendar share. We test whether the observed share differs from this benchmark using a standard proportions z-test.

The same institutional payment cycle that generates PreTOM selling channels reinvestment flows to month-start; when reinvestment buying is unusually strong, the loser rebound overshoots and becomes a crash. Figure 5 displays both dimensions jointly: crash probability is elevated precisely at month-start, while mean WML is highest during PreTOM. Trimming just the worst 1% of days confirms the asymmetry: PreTOM mean WML rises from +9.7 to +17.4 bps (structural), while month-start flips from −4.4 to +5.3 bps (driven entirely by tail events).[14]

What can explain this elevated crash risk at month-start? The institutional cash cycle provides a natural explanation. Funds that sell losers during PreTOM to meet month-end liquidity needs typically reinvest the proceeds at month-start. When reinvestment demand is unusually strong, the surge in buying pressure can cause prices to overshoot, generating a crash. This mechanism is consistent with the observed pattern of crashes concentrating at month-start rather than during PreTOM. It also explains why crashes are not more common during PreTOM: the selling pressure there is driven by liquidity needs, which are relatively stable and predictable, while the reinvestment demand at month-start can be more volatile and subject to sudden surges.

To show this more clearly, we decompose cumulative WML losses during three momentum crash episodes identified following the framework of Daniel and Moskowitz (2016).[15] In every case, PreTOM's share of losses is at or below its calendar share: 29% of −2,204 bps in 2001, 22% of −12,785 bps in 2009. COVID is particularly instructive. The Daniel and Moskowitz (2016) framework requires a prolonged bear market that loads losers with high beta before a market rebound triggers the crash; the 2001 and 2009 episodes fit this pattern. COVID provided no such buildup: the market decline was sudden and followed by an immediate V-shaped recovery, yet momentum crashed severely (Figure 1). Their framework cannot

[14] The Online Appendix shows the full tail-trimming analysis.

[15] August 2001–March 2002, January–September 2009, and February–May 2020 (COVID). The first two episodes fall within the Daniel and Moskowitz (2016) sample; COVID postdates it but exhibits the same momentum crash pattern. The date ranges are ours, chosen to span the full period of sustained momentum losses in each episode.

explain this episode. Ours can: extreme forced selling during PreTOM drove loser prices far below fundamental value, setting up a violent month-start reversal when those positions snapped back. PreTOM earned *positive* returns (+1,424 bps) throughout, while month-start absorbed −6,090 bps (64% of the total loss) as losers surged. No prolonged buildup is needed; the institutional cash cycle generates both the profit and the crash.

5.5 International Evidence

We test whether the PreTOM loser effect extends beyond U.S. equities by constructing daily momentum decile portfolios for 19 developed markets using Compustat Global.[16] Within each country-month, we rank stocks into momentum deciles using cumulative returns over months −12 through −2. Following the U.S. convention, we use size-conditional breakpoints: decile cutoffs are set using stocks above the within-country median market capitalization, then all stocks are assigned to deciles based on those cutoffs. For countries with fewer than 50 large-cap stocks in a given month (Austria, Belgium, Finland, Ireland, New Zealand, Portugal, and Singapore), we pool breakpoints within the broader region. Decile assignments are fixed within each calendar month. Value-weighted portfolio returns use lagged month-end market capitalization as weights. We subtract each country's local market excess return (from Kenneth French's regional factor files) to obtain market-adjusted returns.

Table 10 presents the results. Panel A reports the coefficient on a PreTOM indicator from a pooled regression with country fixed effects and standard errors clustered by country-month. The concentration is driven entirely by the loser leg: losers underperform the local market by 4.5 bps per day during the PreTOM window ($t = -4.60$), while winners show no differential PreTOM concentration (+0.74 bps, $t = 0.78$). The WML difference is 5.3 bps per day ($t = 4.27$), implying that the PreTOM concentration of momentum is not unique to

[16] The 19 countries are: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Hong Kong, Ireland, Italy, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, and the United Kingdom. We exclude Japan, where momentum returns are approximately zero over our sample period, and Canada, for which Compustat Global coverage is limited. The sample begins in 1990 for most countries and in 1993 for Hong Kong and Switzerland, where Compustat coverage is thin in the early 1990s.

U.S. equities.

Panel B reports the PreTOM-minus-rest difference in loser-minus-market returns for selected individual countries. The effect is strongest in Norway (-11.1 bps, $t = -2.82$), followed by the Netherlands (-8.8 bps, $t = -2.05$) and Sweden (-8.5 bps, $t = -2.20$). Hong Kong exhibits the pattern most similar to the U.S.: losers underperform by 6.7 bps per day during PreTOM ($t = -1.61$). Winners show no differential PreTOM concentration in any individual country or in the pooled sample. The effect is absent in Japan, consistent with the well-documented weakness of momentum in that market (Fama and French, 2012; Asness et al., 2013).

The cross-country heterogeneity is substantial. Not every market exhibits a significant PreTOM loser effect individually, which is unsurprising given that single-country portfolio-level tests have limited power. The pooled result, however, is unambiguous: losers underperform during PreTOM in international markets, winners do not, and the pattern mirrors the U.S. finding. A full country-by-country table appears in the Internet Appendix.

5.6 Additional Robustness

Several alternative explanations are ruled out. Mechanical portfolio rebalancing (Harvey et al., 2025) operates at the aggregate level (equities vs. bonds), while our effect is cross-sectional. Passive index rebalancing is ruled out: the window effect is statistically indistinguishable between S&P 500 and non-S&P 500 stocks ($t = -0.82$; Online Appendix Table A5). The pre-month-end window ($T-9$ to $T-4$) does not overlap with the classic turn-of-month window ($T = 0$ through $T+3$); excluding turn-of-month days leaves the Loser $\times$ PreTOM coefficient unchanged. The PreTOM concentration is robust across holding periods from $K = 1$ to $K = 12$ (Online Appendix). After transaction costs following Novy-Marx and Velikov (2016), the PreTOM strategy remains economically significant post-decimalization (Online Appendix Table **??**).

6 Conclusion

The momentum premium accrues almost entirely during six trading days before month-end, driven entirely by losers. The mechanism is institutional dash-for-cash: funds facing month-end liquidity needs disproportionately liquidate past losers during a predictable calendar window. TAQ data confirm the selling pressure spike, the 2024 T+1 settlement reform shifted the precautionary selling window by exactly one day, crash days concentrate at month-start rather than during the profit window, and the pattern extends beyond the U.S. across 19 developed markets. The concentration of the entire premium in six settlement-linked days, while crashes fall elsewhere, is difficult to reconcile with behavioral theories (which require information to arrive on a settlement schedule) or risk-based theories (which require compensation to accrue broadly).

A natural question is whether momentum is special. The approach developed here — decomposing factor returns by day-of-month and asking when anomaly profits actually accrue — can be applied broadly. Many cross-sectional return patterns share a short leg populated by stocks that institutions would rationally sell first under liquidity pressure. If the intra-month structure documented here generalizes, it would suggest that a meaningful fraction of what the literature calls anomaly returns reflects not slow-moving information or priced risk, but the predictable rhythm of institutional cash management.

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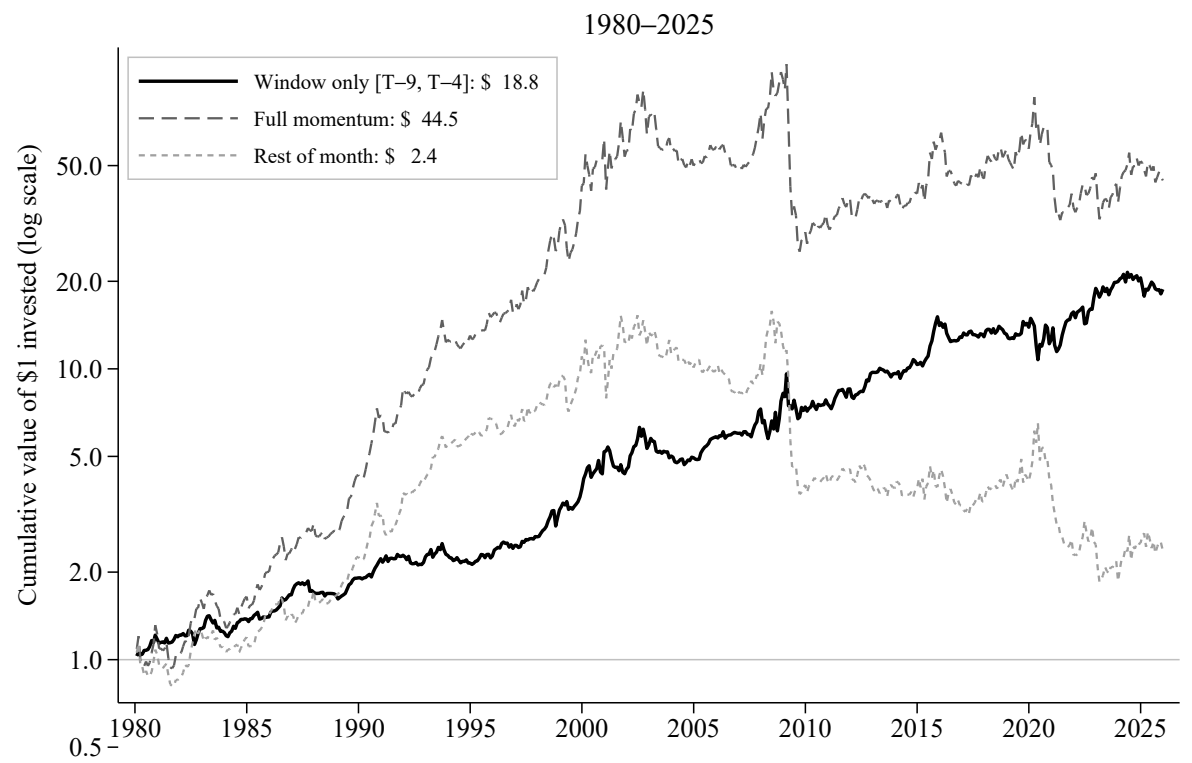

Figure 1: Cumulative wealth from $1 invested in three momentum strategies using fixed monthly decile assignments. WML is the daily winner return (decile 10) minus the loser return (decile 1), both in excess of the risk-free rate. The *Window-only* strategy earns the WML return on days $T-9$ through $T-4$ and earns zero (cash) on all other days. The *Rest-only* strategy earns the WML return on all days *except* $T-9$ through $T-4$ and earns zero during the window. The *Full WML* strategy earns the WML return every day. All three strategies start at $1 on January 2, 1980. Data are daily value-weighted returns using fixed monthly decile assignments from CRSP. Sample: 1980–2025.

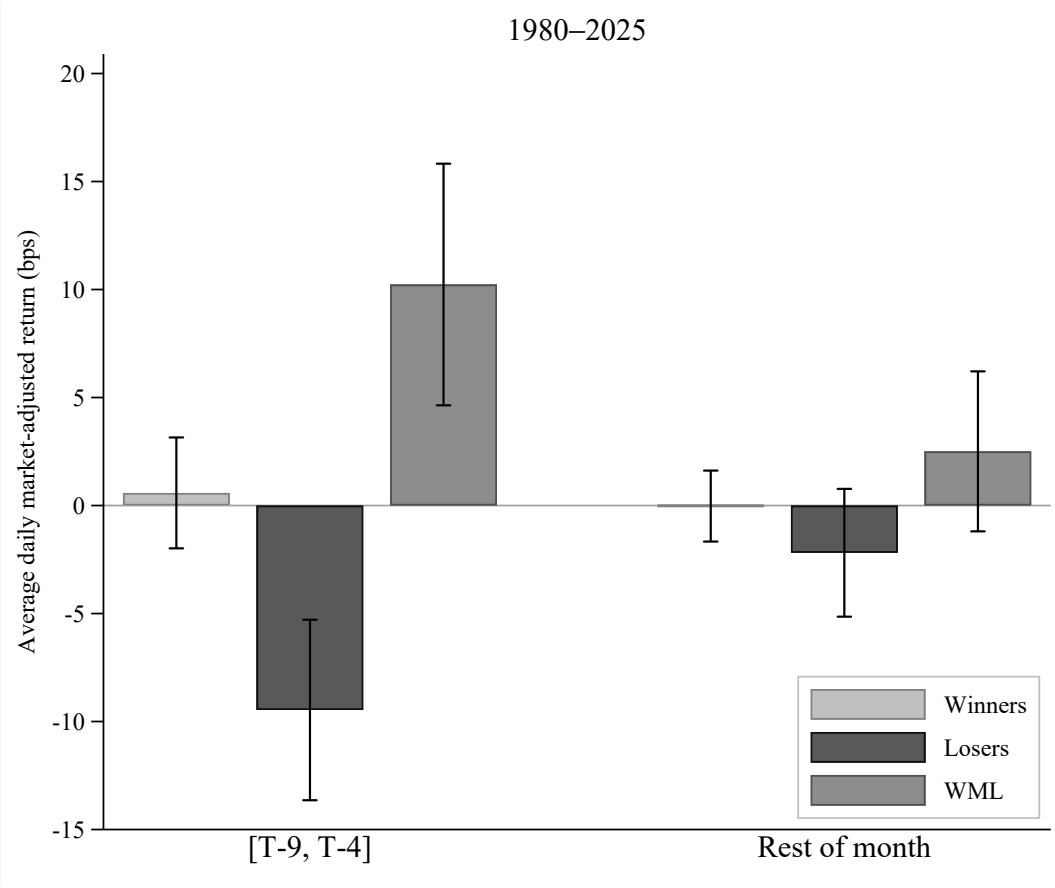

Figure 2: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during the pre-month-end window ($T-9$ to $T-4$) versus the rest of the month. Market-adjusted returns are $r_i - r_{mkt}$ for winners and losers; WML is unchanged. Each bar shows the unconditional time-series mean across all trading days in the indicated window. Whiskers denote 95% confidence intervals. Data are daily value-weighted returns using fixed monthly decile assignments from CRSP. Sample: 1980–2025.

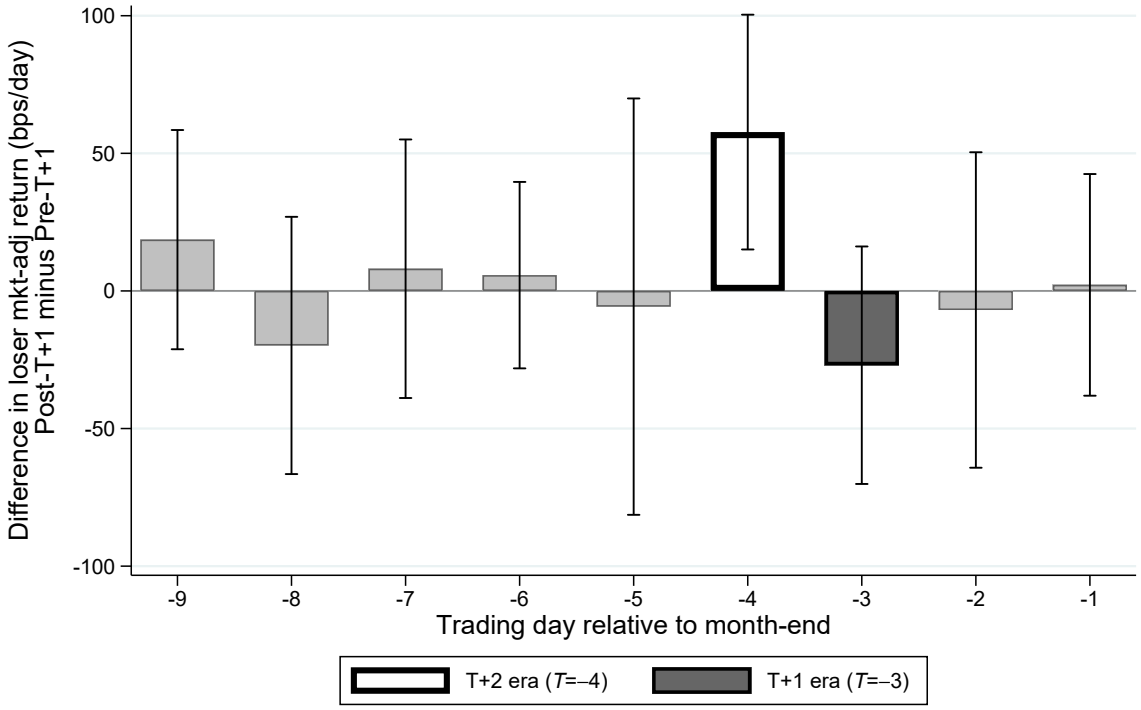

Figure 3: Difference in mean daily market-adjusted VW loser returns (Post-T+1 minus Pre-T+1) at each trading day t, with 95% confidence intervals. The SEC transitioned from T+2 to T+1 settlement on May 28, 2024. Sample: 1980–2025 (533 months pre, 19 months post).

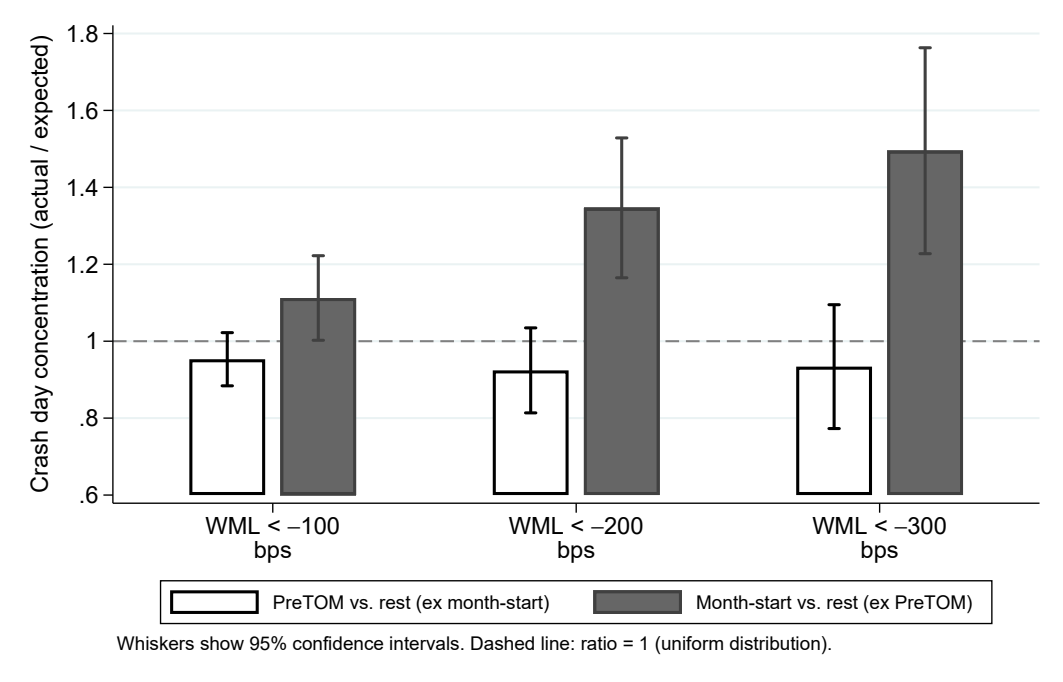

Figure 4: Crash day concentration by calendar window. Each point plots (actual share of crash days in window) / (window's calendar share), with 95% confidence intervals. Each window is compared against remaining trading days only: PreTOM against non-month-start days, and month-start against non-PreTOM days. Under the null that crashes are uniformly distributed, all points equal one (dashed line). Sample: 1980–2025, value-weighted.

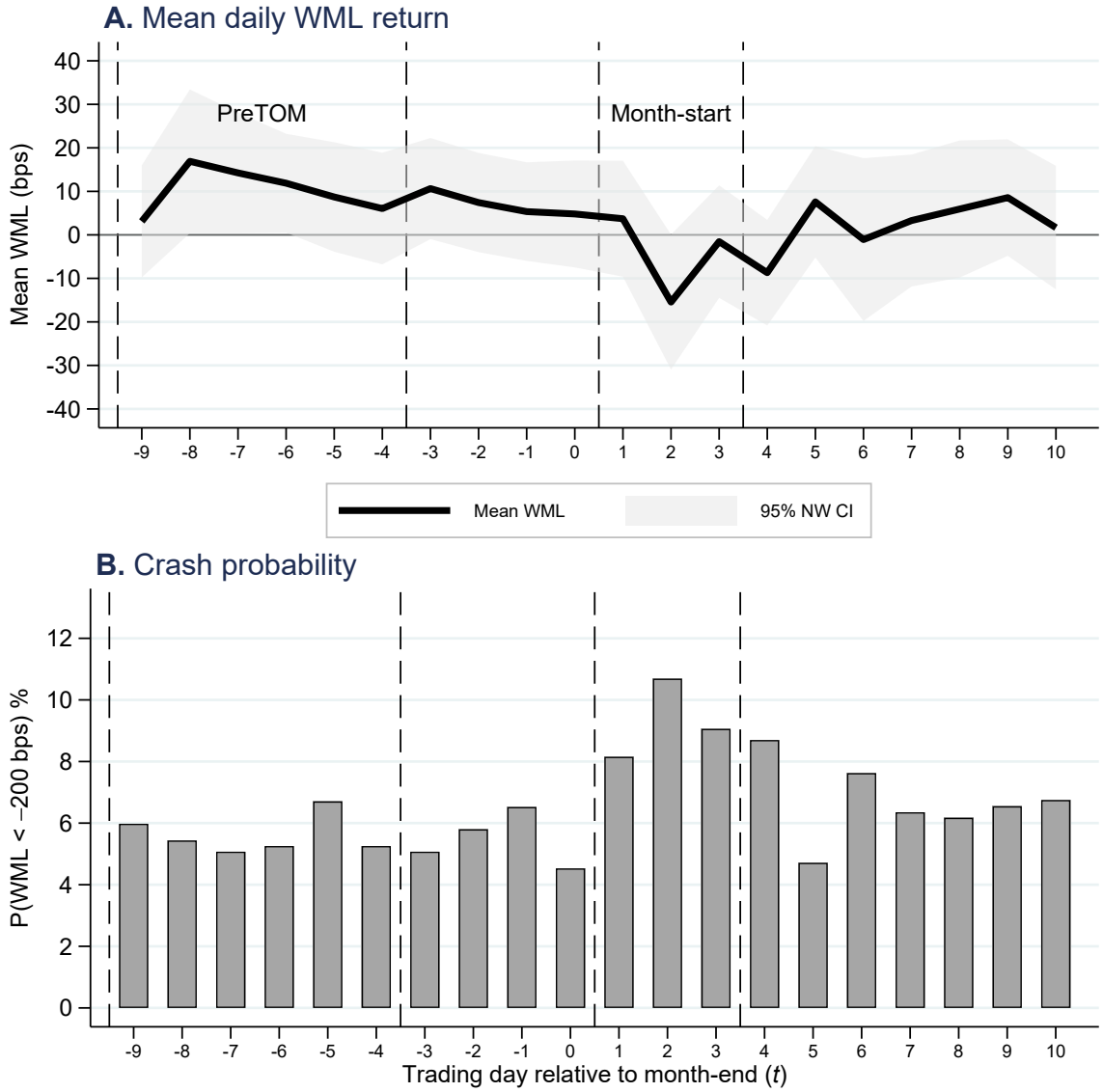

Figure 5: Panel A plots mean daily WML return with Newey-West 95% confidence bands by trading day T relative to month-end. Panel B plots crash probability, defined as Pr(WML $<$ -200 bps). Dashed vertical lines mark the PreTOM window ($T-9$ to $T-4$) and the month-start window ($T+1$ to $T+3$). Sample: 1980–2025, value-weighted.

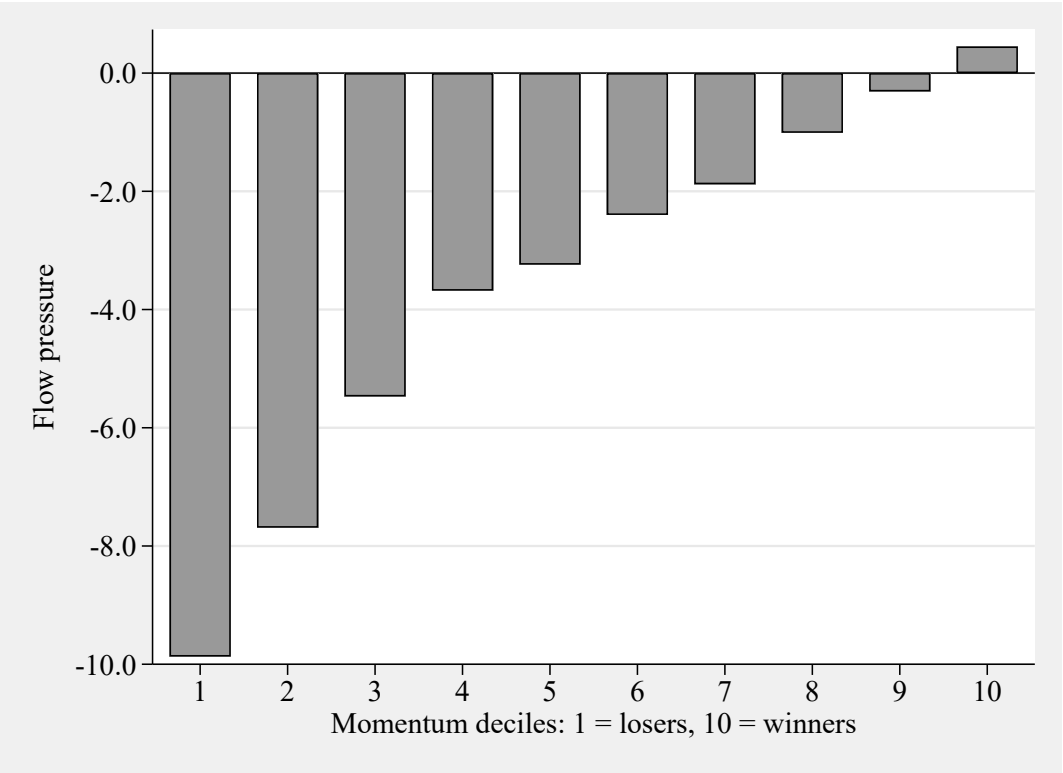

Figure 6: Intra-month net flow pressures by momentum decile. Our sample includes the largest tercile of all stocks in the CRSP (based on their market capitalization in the previous year's last trading day of November), that have a positive daily trading volume, and a share price above \$1. We require that the stock issuer is incorporated in the US. The stocks are sorted to momentum deciles based on their cumulative returns during months $t = -12$ to $t = -1$ (skip-a-month). For each momentum decile we report the sum of mutual fund flow pressures between $T-9$ and $T-4$. Our measure of stock-specific flow pressure is calculated from mutual funds' daily flows assuming that funds invest their flow into stocks in proportion to their holdings at the end of the previous quarter. The flow pressure measures are normalized by the stocks' market capitalizations prevailing one day before the reported flows. The flow pressure measure is the sum of normalized flow pressures multiplied by 100,000 (negative figures corresponding to outflow pressures). Fund flows are those from all mutual funds in the Morningstar and CRSP databases that own equity, that were matched and that have quarterly reported holdings in the CRSP holdings database and daily flows in the Morningstar database. The data encompasses around 30% of all mutual fund total net assets in CRSP and Morningstar. Sample is from beginning of 2010 until end of 2023.

Table 1: Baseline Window Effect on Daily Returns

	(1) Equal Weighted	(2) Value Weighted
Loser	8.022***	2.609**
	(0.717)	(1.279)
Loser × PreTOM	-2.550*	-7.151***
	(1.376)	(2.322)
Constant	5.629***	3.442***
	(0.108)	(0.117)
Observations	53,324,145	53,324,145
Within R^2	0.0000	0.0000

Notes. Dependent variable is daily excess return in basis points. Loser equals one for stocks in the bottom momentum decile. PreTOM equals one during the six-day pre-month-end window (trading days $t-9$ to $t-4$). All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date. VW uses lagged market-capitalization weights within each decile-date cell. Sample: CRSP common stocks, 1980–2025.

Table 2: Window Effect and Bid-Ask Spread

	(1) Equal Weighted	(2) EW + BAS	(3) Value Weighted	(4) VW + BAS
Loser	8.022***	7.161***	2.609**	5.808***
	(0.717)	(0.993)	(1.279)	(1.643)
Loser × BAS		27.635***		−196.989***
		(9.258)		(14.430)
Loser × PreTOM	-2.550*	-5.118***	-7.151***	-8.836***
	(1.376)	(1.747)	(2.322)	(2.992)
Loser × PreTOM × BAS		30.411**		58.771**
		(11.842)		(25.117)
Observations	53,324,145	46,559,900	53,324,145	46,559,900
Within R^2	0.0000	0.0001	0.0000	0.0001

Notes. Dependent variable is daily excess return in basis points. Columns 1 and 3 reproduce the baseline from Table 1 on the full sample. Columns 2 and 4 restrict to the subsample with non-missing bid-ask spreads. BAS is the contemporaneous bid-ask spread as a fraction of the quote midpoint. A positive coefficient on Loser × PreTOM × BAS indicates that the window effect is weaker for illiquid (high-spread) stocks, i.e., concentrated among liquid losers. All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date.

Table 3: Subperiod Stability of the Window Effect

	(1) 1980–2002	(2) 2002–2025
Loser	0.345	6.063***
	(1.292)	(2.081)
Loser × PreTOM	-5.762**	-8.503**
	(2.497)	(3.858)
Observations	30,596,232	22,727,911
Within R^2	0.0000	0.0001

Notes. Dependent variable is daily excess return in basis points. Value-weighted specifications throughout. Sample split at midpoint (July 2002). All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date.

Table 4: Partial Reversal of PreTOM Loser Underperformance

	Full Sample
Loser	1.043
	(1.901)
Loser × PreTOM	-5.585^{**}
	(2.698)
Loser × Post	3.356
	(2.708)
Observations	53,324,145
Within R^2	0.0000

Notes. Dependent variable is daily excess return in basis points. Loser equals one for bottom-decile momentum stocks (fixed monthly sorting). PreTOM equals one during trading days $t-9$ to $t-4$ relative to month-end ($t = 0$). Post equals one during trading days $t-3$ to $t+3$ surrounding month-end. The positive but insignificant Post coefficient is consistent with partial price recovery following the PreTOM selling pressure. Value-weighted (lagged market cap). Firm and date fixed effects. Standard errors (in parentheses) clustered by firm and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Window Effect in Non-Quarter-End Months

	(1) Non-QE, Value Weighted
Loser	4.132***
	(1.539)
Loser × PreTOM	-7.523***
	(2.828)
Constant	4.636***
	(0.137)
Observations	35,417,562
Within R^2	0.0000

Notes. Dependent variable is daily excess return in basis points. Value-weighted. Sample restricted to non-quarter-end months (Jan, Feb, Apr, May, Jul, Aug, Oct, Nov). All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date.

Table 6: Selling Pressure During PreTOM: Firm and Date Fixed Effects

	Net Sell Pressure (1)	Sell Share (2)
Loser	0.0122***	0.0061***
	(0.0006)	(0.0003)
Loser × PreTOM	0.0020**	0.0010**
	(0.0008)	(0.0004)
Fixed effects	Firm, Date	Firm, Date
Observations	17,861,276	17,861,276
Adj. R^2	0.0361	0.0361

Notes. OLS regressions of daily selling pressure on loser status and its interaction with the PreTOM window ($T \in [-9, -4]$). Net sell pressure = (sell volume − buy volume) / total volume; sell share = sell volume / total volume. Trade direction assigned using Lee-Ready (WRDS Intraday Indicators). Sample: 2003–2022, approximately 17.9 million stock-day observations (extracted from the main CRSP panel). Firm and date fixed effects included. Standard errors double-clustered by firm and date in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: T+1 Settlement Difference-in-Differences

	(1) T+2 baseline (Sep 2017–2025)	(2) Full baseline (1980–2025)
Panel A: Cell Means (loser market-adjusted return, bps/day)		
	Pre-T+1 / Post-T+1	
Early-month $[T{+}5, T{+}8]$ (control)	-5.1 / $+5.0$	
$T{-}3$	-5.2 / -32.2	
$T{-}4$	-8.4 / $+49.4$	
Panel B: Portfolio-Level DiD (early-month control)		
	(1) T+2 baseline (Sep 2017–2025)	(2) Full baseline (1980–2025)
$\mathbf{1}[T{=}{-}4]$	-21.34	-3.29
	(19.94)	(5.30)
$\mathbf{1}[T{=}{-}3]$	-19.02	-0.14
	(19.05)	(5.46)
Post T+1	$+6.88$	$+10.13$
	(17.49)	(14.17)
β_4: $\mathbf{1}[T{=}{-}4] \times$ Post T+1	$+65.65^{**}$	$+47.60^{*}$
	(32.12)	(25.54)
β_5: $\mathbf{1}[T{=}{-}3] \times$ Post T+1	-18.24	-37.11
	(31.74)	(25.77)
$\beta_4 - \beta_5$ (full shift)	$+83.89^{**}$	$+84.72^{***}$
	(37.79)	(30.24)
Observations	300	1,656

Notes: DiD test of the T+2→T+1 settlement transition (May 28, 2024). Dependent variable: daily market-adjusted VW loser return (bps). Sample: days $T{-}4$, $T{-}3$, $T{-}2$, with early-month control as control. Panel A shows cell means. Panel B reports the portfolio-level unified DiD with robust standard errors. β_4 tests whether $T{-}4$ was relieved after T+1. β_5 tests whether $T{-}3$ absorbed selling pressure. $\beta_4 - \beta_5$ measures the full shift. Column 1 restricts the pre-period to the T+2 era (Sep 2017–May 2024). Column 2 uses the full baseline (1980+). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: T+1 Settlement Difference-in-Differences (continued)

Panel C: Stock-Level Triple-Difference (early-month control, firm + date FE, VW)	Full baseline (1980–2025)
β_4: Loser $\times$ $\mathbf{1}[T{=}{-4}]$ $\times$ Post T+1	$+161.45^{**}$
	(65.89)
β_5: Loser $\times$ $\mathbf{1}[T{=}{-3}]$ $\times$ Post T+1	-43.51
	(57.08)
$\beta_4 - \beta_5$ (full shift)	$+204.96^{**}$
	(80.58)
Observations	4,915,026

Notes: Stock-level triple-difference for the T+2→T+1 settlement transition. Dependent variable: daily excess return (bps). Loser $\times$ day $\times$ PostT1 interactions with firm and date fixed effects, value-weighted, standard errors clustered by date. See Table 7 for details. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Dividend Income and PreTOM Selling Pressure

Panel A: Dividend Payer Composition		
	D1 Losers	D10 Winners
Non-payer	88.4%	72.0%
Quarterly payer	8.7%	23.2%
Other (semi-annual, annual, monthly)	2.9%	4.8%
Panel B: Within-Quarter Returns (Quarterly Payers Only)		
	VW	
Loser × PreTOM × Month 1 (just paid)	−6.28	
	(−1.32)	
Loser × PreTOM × Month 2 (1 month after)	-10.56^{**}	
	(−1.98)	
Loser × PreTOM × Month 3 (about to pay)	−7.74	
	(−1.57)	
N	2,450,000	
Panel C: Within-Quarter TAQ Selling Pressure (Quarterly Payers Only)		
	Net sell pressure	Sell share
PreTOM × Month 2 (1 month after)	$+0.0038^{***}$	$+0.0019^{***}$
	(4.99)	(4.99)
PreTOM × Month 3 (about to pay)	$+0.0018^{**}$	$+0.0009^{**}$
	(2.34)	(2.34)
N	6,200,000	6,200,000

Notes: Panel A reports the share of stock-month observations by dividend frequency classification (CRSP distributions, modal frequency). Panels B and C restrict to quarterly dividend payers within D1 and D10 momentum deciles. Month 1 = the month in which the quarterly dividend was paid (or the month immediately after); Month 2 = one month after payment; Month 3 = about to pay again. Panel B reports stock-level regressions of daily excess returns (basis points) on Loser × PreTOM interacted with month-within-quarter indicators, with firm and date fixed effects, value-weighted by lagged market capitalization. Panel C reports TAQ net selling pressure and sell share regressions on PreTOM × month-within-quarter indicators, with firm and date fixed effects; all quarterly payers included. Standard errors double-clustered by firm and date. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: PreTOM Returns by Loser Freshness

	Coefficient	t-statistic	p-value
Panel A: Tercile split within D1 losers			
PreTOM × Fresh (bottom tercile $\text{ret}_{3,1}$)	−3.30	−2.17	0.030
PreTOM × Stale (top tercile $\text{ret}_{3,1}$)	+1.24	1.02	0.308
Test Fresh = Stale (F-statistic)	$F = 7.15$, $p = 0.008$		
Panel B: Continuous freshness			
Without BAS controls (N = 9,954,220)			
PreTOM × fresh_z	−1.51	−2.10	0.036
With BAS controls (N = 9,029,730)			
PreTOM × fresh_z	−1.77	−2.38	0.017
PreTOM × BAS	+33.1	+2.30	0.021
December falsification (with BAS controls)			
PreTOM × fresh_z (non-December)	−2.07	−2.72	0.007
December × PreTOM × fresh_z	+3.77	+1.61	0.107
Total December effect	+1.70	+0.74	0.459

Notes: Sample consists of bottom-decile (D1) momentum losers only, 1980–2025. Freshness is measured by $\text{ret}_{3,1}$ (cumulative return over months −3 through −1), standardized within D1 each month and sign-flipped so that higher values indicate worse recent performance. In Panel A, D1 is split into terciles by freshness; PreTOM × Fresh and PreTOM × Stale are the window effects for the bottom and top terciles. Panel B uses the continuous standardized measure. All specifications include firm and date fixed effects; standard errors are double-clustered by firm and date. Returns in basis points per day.

Table 10: International Evidence: PreTOM Returns across Developed Markets

	PreTOM − Rest	t-statistic
Panel A: Pooled regression (19 countries, 1990–2025)		
Losers − Market	−4.47	(−4.60)
Winners − Market	+0.74	(0.78)
WML	+5.25	(4.27)
Panel B: Selected countries — Losers − Market (bps/day)		
Norway	−11.10	(−2.82)
Netherlands	−8.80	(−2.05)
Sweden	−8.51	(−2.20)
Switzerland	−7.03	(−1.42)
Belgium	−6.93	(−1.71)
Hong Kong	−6.68	(−1.61)
UK	−5.28	(−1.40)
France	−4.49	(−1.28)
Germany	−3.75	(−1.11)
Australia	−2.50	(−0.81)

Notes: Daily value-weighted portfolio returns from Compustat Global, 1990–2025 (HKG/CHE from 1993). Nineteen developed markets excluding Japan and Canada. Momentum deciles are formed within each country-month using cumulative returns over months −12 through −2. Decile breakpoints are set using stocks above the within-country median market capitalization (analogous to NYSE breakpoints in the U.S. analysis); all stocks are then assigned to deciles based on those cutoffs. For countries with fewer than 50 qualifying stocks, breakpoints are pooled within the broader region. Returns are market-adjusted using Kenneth French's regional factor files. PreTOM denotes trading days $T-9$ through $T-4$ relative to month-end. Panel A reports the coefficient on a PreTOM indicator from a pooled regression with country fixed effects; standard errors are clustered by country-month. Panel B reports the mean difference (PreTOM minus rest of month) in daily loser-minus-market returns for the ten largest markets; t-statistics are clustered by month. The winner difference is insignificant in every country and pooled ($t = 0.78$).

Appendix

Table A1: Quarter-End Amplification of the Window Effect

	(1) Value Weighted
Loser	4.039***
	(1.544)
Loser × PreTOM	-7.496***
	(2.828)
Loser × PreTOM × QtrEnd	0.990
	(5.369)
Loser × QtrEnd	-4.249
	(2.867)
Constant	3.442***
	(0.115)
Observations	53,324,145
Within R^2	0.0000

Notes. Dependent variable is daily excess return in basis points. Value-weighted. QtrEnd equals one in March, June, September, December. A significant Loser × PreTOM × QtrEnd would indicate amplification at quarter-ends, consistent with window dressing. All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date.

Table A2: T+1 Settlement: Falsification Tests

Test	DiD	t-stat	p-value	Prediction
Real: Losers, $T-4$ vs. $T-3$, May 2024	**+69.45**	**2.36**	**0.018**	**Nonzero**
Placebo days (losers, May 2024):				
$T-6$ vs. $T-7$	+6.63	0.24	0.810	Zero
Placebo dates (losers, $T-4$ vs. $T-3$):				
May 28, 2020	−0.39	−0.01	0.990	Zero
May 28, 2018	−1.89	−0.08	0.933	Zero

Notes: Falsification tests for the T+1 settlement DiD (Table 7). All tests use the same specification: market-adjusted VW loser returns regressed on a day indicator, a post-event indicator, and their interaction, with robust standard errors. Placebo days shift the comparison to $T-6$ vs. $T-7$, two days interior to both the T+2 and T+1 selling windows where no shift is predicted. Placebo dates use fake event dates when no regulatory change occurred. All falsification interactions are small and statistically insignificant. Sample: 1980–2025.

Table A3: T+3→T+2 Settlement Deadline-Day Difference-in-Differences

Panel A: 2×2 Cell Means

	Pre-T+2 (1980–Aug 2017)	Post-T+2 (Sep 2017–2025)	Diff
$T-3$	-2.1 ($n = 452$)	-19.2 ($n = 100$)	-17.1
$T-4$	-4.9 ($n = 452$)	-18.0 ($n = 100$)	-13.1
Diff ($T-4$ minus $T-3$)	-2.8	$+1.2$	$+\mathbf{4.0}$

Panel B: DiD Regression

	Coefficient	Robust SE	t-stat	p-value
$\mathbf{1}[t{=}-4]$	-2.82	6.61	-0.43	0.670
Post T+2	-17.08	14.83	-1.15	0.250
$\mathbf{1}[t{=}-4] \times$ Post T+2	$+4.02$	21.06	0.19	0.849
Constant	-2.09	4.93	-0.42	0.671
Observations		1,104		
R^2		0.0030		

Notes: Deadline-day difference-in-differences test of the T+3 to T+2 settlement transition (September 5, 2017). Dependent variable: daily market-adjusted VW loser portfolio return (bps). Sample restricted to trading days $t = -4$ and $t = -3$ only, the same pair used in the T+1 test (Table 7). The DiD is near zero and insignificant, as expected: the 2017 reform reduced the settlement mismatch but did not eliminate it, so the $T-4/T-3$ pair did not diverge. Robust standard errors. Sample: 1980–2025.

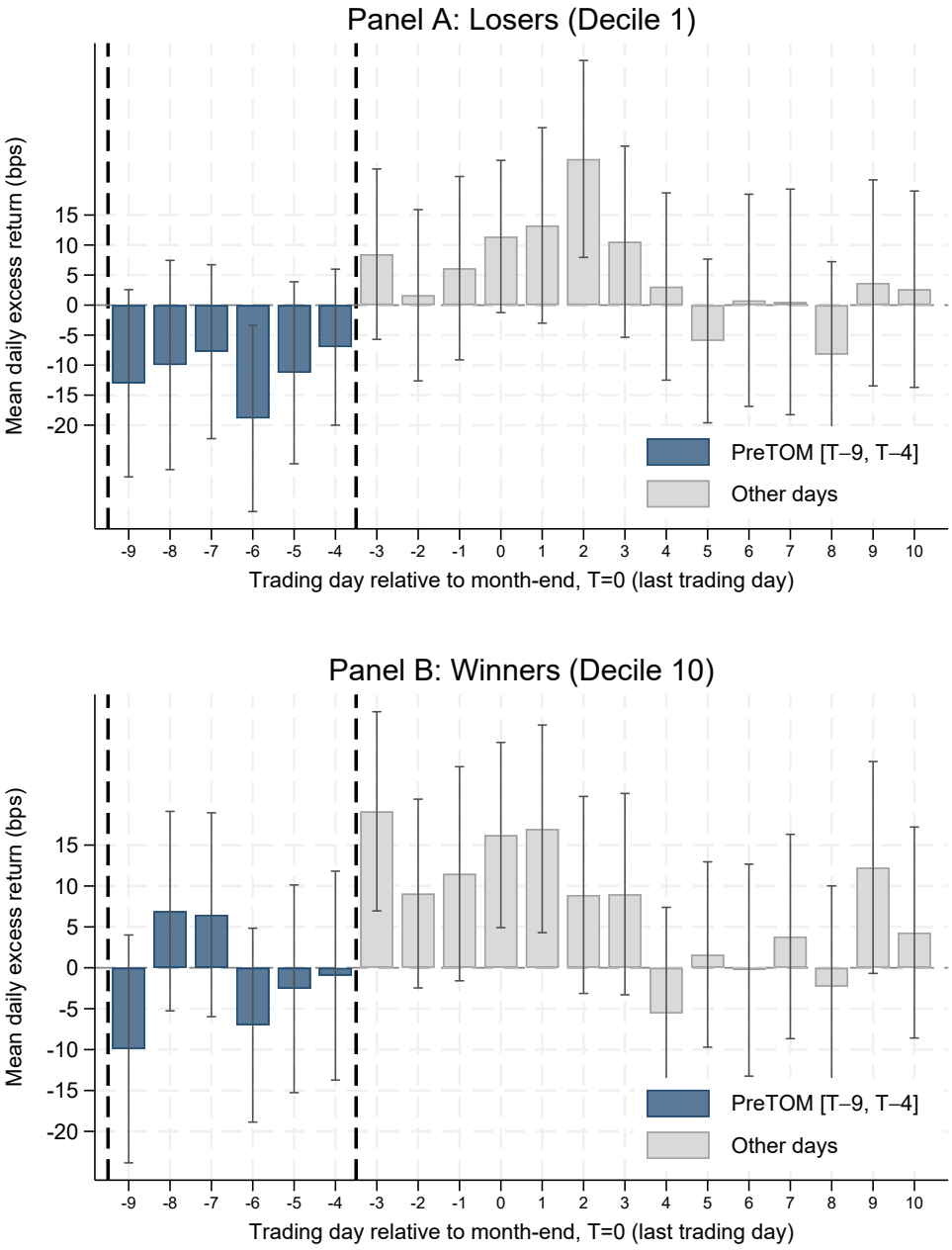

Figure A1: Mean daily excess returns (basis points) of loser and winner decile portfolios by trading day t relative to month-end. Panel A: loser portfolio (decile 1). Panel B: winners (decile 10). Bars corresponding to the PreTOM window ($T-9$ through $T-4$) are highlighted. Data are value-weighted daily returns using fixed monthly decile assignments from CRSP. Sample: 1980–2025.

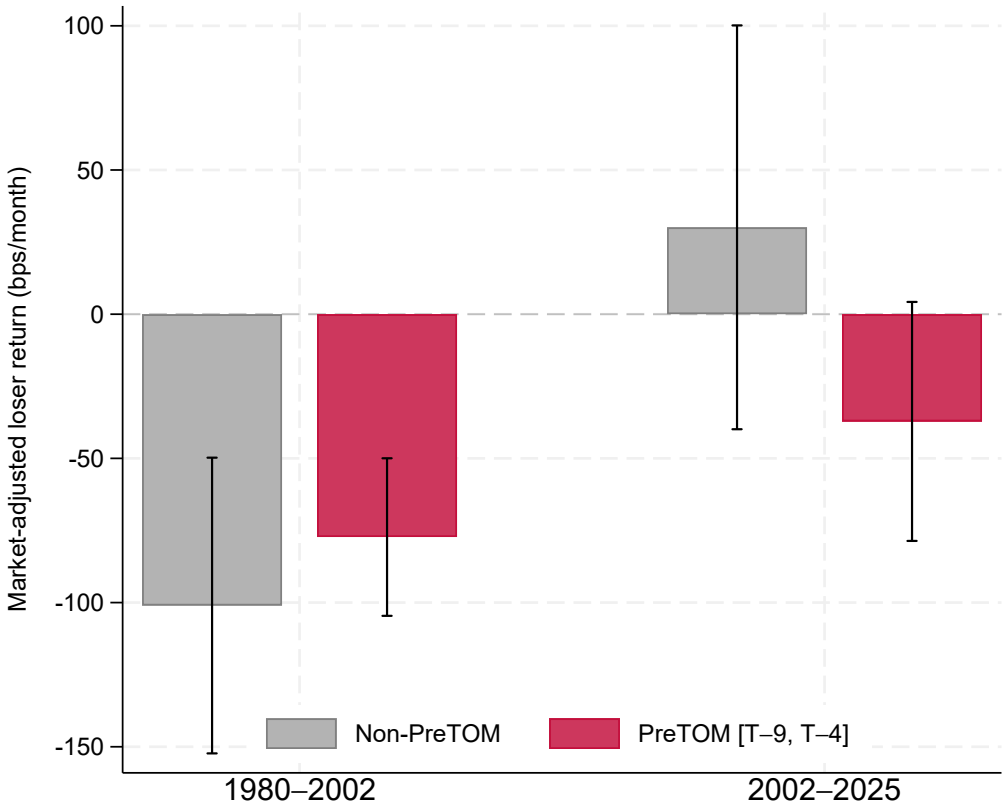

Figure A2: Market-adjusted monthly loser returns during the PreTOM window versus the rest of the month, by subperiod. Daily market-adjusted loser returns ($r_i - r_{mkt}$) are compounded separately within the PreTOM window (days $T-9$ to $T-4$) and the non-PreTOM window (all other trading days). Bars show the time-series mean of monthly compounded returns (basis points); whiskers denote 95% confidence intervals. Data are value-weighted returns using fixed monthly decile assignments from CRSP.

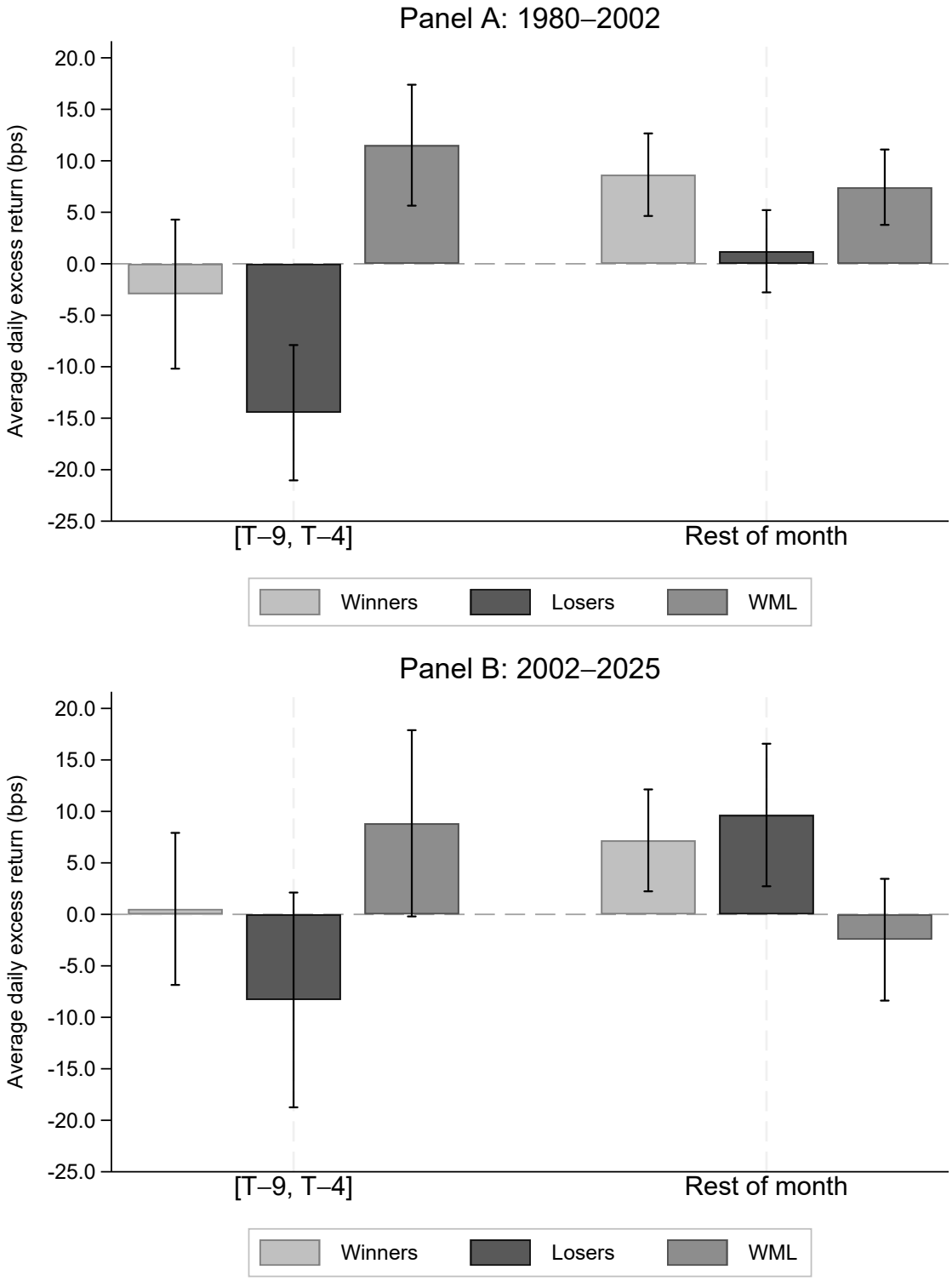

Figure A3: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during the pre-month-end window ($T-9$ to $T-4$) versus the rest of the month, by subperiod. Market-adjusted returns are $r_i - r_{mkt}$ for winners and losers; WML is unchanged. Bars show time-series means; whiskers denote 95% confidence intervals. Data are daily value-weighted returns using fixed monthly decile assignments from CRSP.

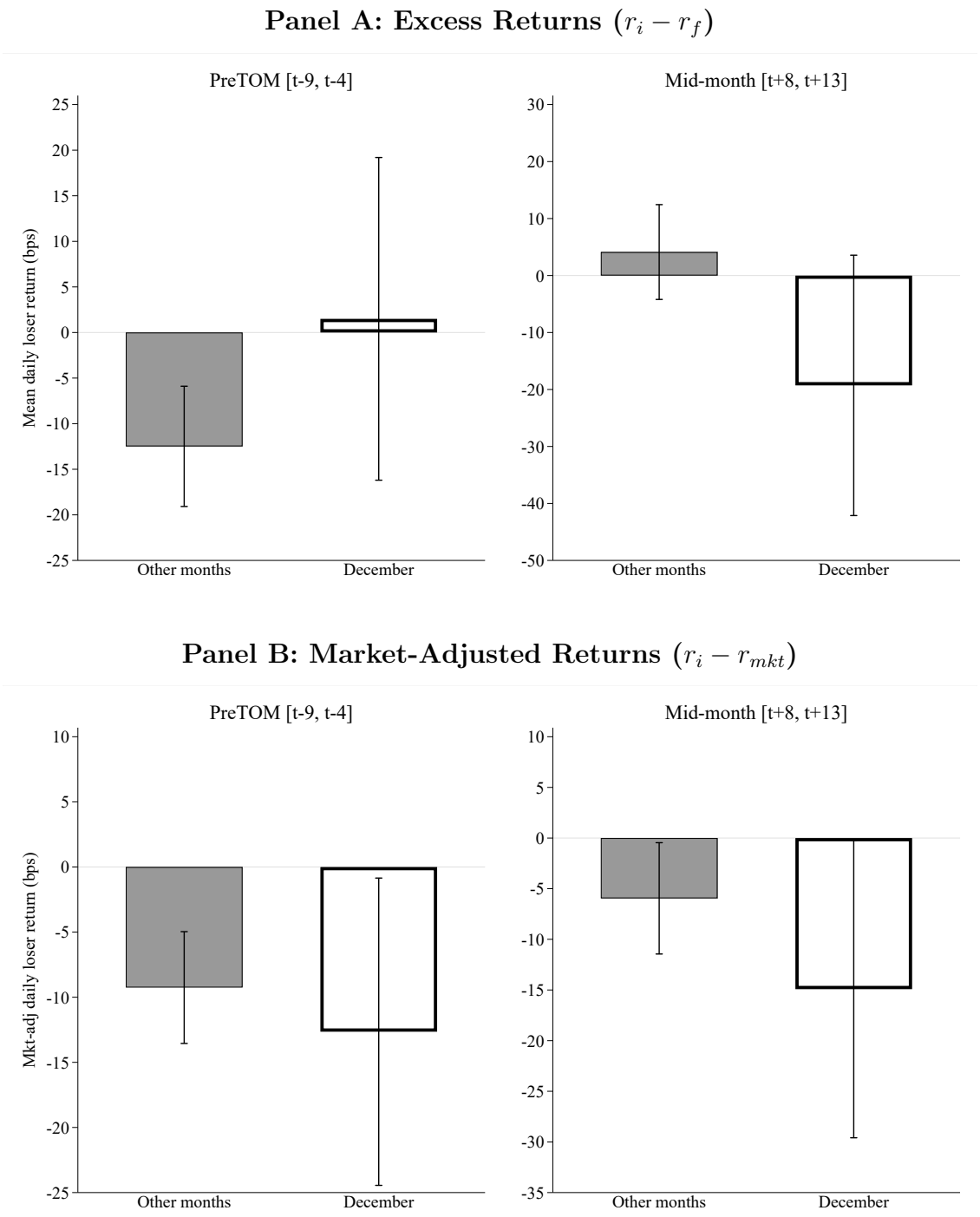

Figure A4: Mean daily loser returns during the pre-month-end window ($T-9$ to $T-4$, left) and the first two weeks of the month (right) in December versus all other months. Panel A uses excess returns ($r_i - r_f$); Panel B uses market-adjusted returns ($r_i - r_{mkt}$). Dots show time-series means; whiskers denote 95% confidence intervals. Value-weighted returns. Sample: 1980–2025.

Table A4: Winner Window Effect on Daily Returns

	EW	VW
Winner	-0.23^{***}	0.04
	(0.07)	(0.10)
Winner × PreTOM	-1.27	1.55
	(1.55)	(2.03)
Fixed effects	Firm, Date	Firm, Date
Observations	53,324,149	53,324,149
Within R^2	0.0000	0.0000

Notes. Dependent variable is daily excess return in basis points. Value-weighted (VW) regressions use lagged market-capitalization weights. Standard errors (in parentheses) are double-clustered by firm and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Window Effect and S&P 500 Membership

	Non-S&P 500	S&P 500	Full Sample
	(1)	(2)	(3)
Loser	1.806*	4.018**	2.429**
	(0.980)	(1.803)	(1.103)
Loser × PreTOM	−4.622**	−8.769***	−5.916**
	(1.825)	(3.237)	(2.307)
Loser × S&P 500			0.408
			(1.603)
Loser × PreTOM × S&P 500			−2.520
			(3.060)
Observations	47,863,849	5,460,280	53,324,145
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable is daily excess return in basis points. Value-weighted using lagged market-capitalization weights. Columns (1) and (2) restrict the sample to non-S&P 500 and S&P 500 constituent stocks, respectively. Column (3) estimates the full sample with an S&P 500 interaction; an insignificant Loser × PreTOM × S&P 500 coefficient indicates that the window effect does not differ across index membership. All specifications include firm and date fixed effects. Standard errors (in parentheses) are two-way clustered by firm and date. Sample: 1980–2025.